

The LisbOn KInetics Boltzmann solver

*** discretization in a non-uniform energy grid ***

Future version: LoKI-B_v3.0.0

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March 23, 2025

1 The two-term electron Boltzmann equation

The two-term electron Boltzmann equation (EBE) including a non-conservative collisional operator can be written under different frameworks (with $\gamma \equiv \sqrt{2e/m_e}$)¹:

- considering a **temporal growth model**, which assumes an exponential temporal growth of the electron density with a growth constant $\langle \nu_{\text{eff}} \rangle \equiv \langle \nu_{\text{ion}} \rangle - \langle \nu_{\text{att}} \rangle \equiv \sum_{k,i} \delta_{k_i} [\langle \nu_{k_i, \text{ion}} \rangle - \langle \nu_{k_i, \text{att}} \rangle]$ corresponding to the electron net creation frequency for the system under study $\partial n_e(t)/\partial t = \langle \nu_{\text{eff}} \rangle n_e(t)$, with $\langle \nu_{\text{ion}} \rangle$ and $\langle \nu_{\text{att}} \rangle$ the corresponding electron mean frequencies for ionisation and attachment, respectively, and where the sums extend over all levels of all gases in the mixture:

$$\frac{1}{\gamma\sqrt{u}} \frac{\langle \nu_{\text{eff}} \rangle}{N} u f(u) + \frac{1}{N} \frac{1}{\gamma} \frac{dG(u)}{du} = S(u) \quad (2a)$$

$$f^1(u) = -\frac{\zeta(E/N)}{\Omega_{\text{PT}}(u)} \frac{df(u)}{du} \quad ; \quad (2b)$$

- considering a **spatial growth model**, which assumes an exponential spatial growth of the electron density in the direction opposite to that of the applied electric field, with a growth constant $\alpha_{\text{eff}} \equiv \alpha - \eta \equiv \sum_{k,i} \delta_{k_i} (\alpha_{k_i} - \eta_{k_i})$ corresponding to the effective electron Townsend coefficient $\partial n_e(z)/\partial z = \alpha_{\text{eff}} n_e(z)$, with α and η the corresponding first and second Townsend coefficients for ionisation and attachment, respectively:

$$\frac{1}{3} \frac{\alpha_{\text{eff}}}{N} u f^1(u) + \frac{1}{N} \frac{1}{\gamma} \frac{dG(u)}{du} = S(u) \quad (3a)$$

$$f^1(u) = -\frac{(\alpha_{\text{eff}}/N)}{\Omega_{\text{SST}}(u)} f(u) - \frac{(E/N)}{\Omega_{\text{SST}}(u)} \frac{df(u)}{du} \quad . \quad (3b)$$

- considering a **time-dependent** solution under the action of a non-oscillatory ($\omega = 0$) electric-field along

¹In a HF case the quantity $f^1(u)$ in (2b) represents the *real part* of the first-anisotropy of the electron distribution function. Indeed, in this case $f^1(u, t) = \tilde{f}^1(u) e^{j\omega t}$ with

$$\tilde{f}^1(u) = -\gamma\sqrt{u} \frac{E_p}{N\gamma\sqrt{u}\Omega_c(u) + j\omega} \frac{df(u)}{du} = -\frac{\zeta(E/N)}{\Omega_{\text{PT}}(u)} \frac{df(u)}{du} + j \frac{\zeta(E/N)}{\Omega_{\text{PT}}(u)} \frac{\frac{\omega/N}{\Omega_c\gamma\sqrt{u}} df(u)}{du} \quad . \quad (1)$$

the z -axis, assuming a space-independent exponential temporal growth of the electron density:

$$\frac{1}{N} \frac{1}{\gamma} \sqrt{u} \frac{\partial f(u, t)}{\partial t} + \frac{1}{\gamma \sqrt{u}} \frac{\langle \nu_{\text{eff}} \rangle(t)}{N} u f(u, t) + \frac{1}{N} \frac{1}{\gamma} \frac{\partial G(u, t)}{\partial u} = S(u, t) \quad (4a)$$

$$f^1(u, t) = - \frac{(E(t)/N)}{\Omega_c(u, t)} \frac{\partial f(u, t)}{\partial u} \quad . \quad (4b)$$

In these equations:

- $f_0 \equiv f$ is the isotropic part of the electron distribution function, identified with the electron energy distribution function (EEDF) and satisfying the normalisation condition $\int_0^\infty f(u) \sqrt{u} du = 1$; and f_1 is the (real part of the) first-anisotropy of the electron distribution function;
- $u = m_e v^2 / (2e)$ is the electron kinetic energy in eV (with m_e and e the electron mass and charge, respectively);
- the quantities Ω_{PT} , Ω_{ST} and Ω_c have the dimensions of a cross section, being defined as

$$\Omega_{\text{PT}}(u) \equiv \Omega_c(u) + \frac{[1/(\gamma^2 u)] (\omega/N)^2}{\Omega_c(u)} \quad (5a)$$

$$\Omega_{\text{ST}}(u) \equiv \sigma_c(u) \quad (5b)$$

$$\Omega_c(u, t) \equiv \sigma_c(u) + \frac{1}{\gamma \sqrt{u}} \frac{1}{N n_e(t)} \frac{dn_e(t)}{dt} = \sigma_c(u) + \frac{1}{\gamma \sqrt{u}} \frac{\langle \nu_{\text{eff}} \rangle(t)}{N} \quad , \quad (5c)$$

where $\sigma_c(u) \equiv \sum_k \chi_k \sigma_{k,c}$ is the electron-neutral total momentum-transfer cross section for the system under study, $\chi_k \equiv N_k/N$ the fraction of gas k , and $\sigma_{k,c}$ the corresponding cross section for gas k .

Here, the time-dependent collision frequency $\Omega_c(u, t)$ accounts for the intrinsic time evolution of the electron density, and the same expression (5c) can be used to define the time-independent collision frequency $\Omega_c(u)$ by simply removing the t -dependence;

- $G \equiv G_E + G_{\text{el}} + G_{\text{CAR}} + G_{ee}$ is the *upflux* function that contains power gain/loss contributions due to, in order, the applied electric-field, the elastic collisions, the rotational collisions (in the continuous approximation for rotations, CAR, with a Chapman-Cowling correction due to the gas temperature written here for the case of homonuclear diatomic molecules), and the electron-electron collisions, respectively

$$G_E(u) = N \gamma \frac{u}{3} \frac{1}{\zeta} \frac{E}{N} f^1(u) \quad (6a)$$

$$G_{\text{el}}(u) = - \sum_k 2 \frac{m_e}{M_k} \nu_{k,c}^{\text{el}}(u) u^{3/2} \left[f(u) + \frac{k_B T_g}{e} \frac{df(u)}{du} \right] \quad (6b)$$

$$G_{\text{CAR}}(u) = - \sum_k 4 B_k \nu_{0,k} u^{1/2} \left[f(u) + \frac{k_B T_g}{e} \frac{df(u)}{du} \right] \quad (6c)$$

$$G_{ee}(u) = -2 \nu_{ee}(u) u^{3/2} \left[I(u) f(u) + J(u) \frac{df(u)}{du} \right] \quad , \quad (6d)$$

where $\nu_{k,c}^{\text{el}} \equiv N_k \gamma \sqrt{u} \sigma_{k,c}^{\text{el}}$ is the electron-neutral elastic momentum-transfer collision frequency for the gas k ($\sigma_{k,c}^{\text{el}}$ being the corresponding cross section), $\nu_{0,k} \equiv N_k \gamma \sqrt{u} \sigma_{0,k}$ is the electron-neutral collision frequency for the rotational excitation of gas k ($\sigma_{0,k} \equiv 8\pi Q_k^2 a_0^2 / 15$ being the corresponding cross section, with a_0 the Bohr radius), B_k and Q_k are the rotational constant and the quadrupole-moment constant (in units of ea_0^2) of the gas k , respectively, $I(u)$ and $J(u)$ are the relevant Spitzer integrals, and $\nu_{ee} \equiv 4\pi [e^2 / (4\pi \epsilon_0 m_e)]^2 \ln \Lambda_c / v^3 n_e$ is the electron-electron collision frequency (with ϵ_0 the vacuum permittivity, $\ln \Lambda_c$ the Coulomb logarithm ($\Lambda_c \equiv 12\pi n_e \lambda_D^3$) and $\lambda_D = \sqrt{\epsilon_0 k_B T_e / (n_e e^2)}$ the Debye length);

- $S \equiv \sum_{i,j>i} S_{i,j} + \sum_i S_{i,\text{ion}} + \sum_i S_{i,\text{att}}$ is the discrete collision operator that contains power loss/gain contributions due to inelastic/superelastic mechanisms, ionisation and attachment (recombination is not

yet considered), respectively

$$\begin{aligned} S_{i,j}(u) &= \delta_i [(u + V_{i,j})\sigma_{i,j}(u + V_{i,j})f(u + V_{i,j}) - u\sigma_{i,j}(u)f(u)] \\ &+ \delta_j \frac{g_i}{g_j} [u\sigma_{i,j}(u)f(u - V_{i,j}) - (u + V_{i,j})\sigma_{i,j}(u + V_{i,j})f(u)] \end{aligned} \quad (7a)$$

$$\begin{aligned} S_{i,\text{ion}}(u) &= \delta_i \left\{ \int_{u+V_{i,\text{ion}}}^{2u+V_{i,\text{ion}}} u' \sigma_{i,\text{ion}}^{\text{sec}}(u', u' - V_{i,\text{ion}} - u) f(u') du' \right. \\ &+ \left. \int_{2u+V_{i,\text{ion}}}^{\infty} u' \sigma_{i,\text{ion}}^{\text{sec}}(u', u) f(u') du' - u\sigma_{i,\text{ion}}(u)f(u) \right\} \end{aligned} \quad (7b)$$

$$S_{i,\text{att}}(u) = -\delta_i u\sigma_{i,\text{att}}(u)f(u) \quad . \quad (7c)$$

In (7a)-(7c), the subscripts i, j and the densities δ_i, δ_j refer to any state of any gas in the system; $\sigma_{i,j}$ is the electron-collision cross section for the $i \rightarrow j$ excitation with energy threshold $V_{i,j}$ and statistical weights g_i and g_j , respectively; $\sigma_{i,\text{att}}(u)$ is the electron-collision cross section for the attachment of state i ; $\sigma_{i,\text{ion}}^{\text{sec}}(u', u)$ is the single differential cross section (SDCS) for the ionisation of state i with energy threshold $V_{i,\text{ion}}$; and

$$\sigma_{i,\text{ion}}(u) = \int_0^{(u-V_{i,\text{ion}})/2} \sigma_{i,\text{ion}}^{\text{sec}}(u, u') du' \quad (8)$$

is the integral cross section for the ionisation of state i .

Ionisation can be accounted for using other descriptions, namely:

- **oneTakesAll**

$$S_{i,\text{ion}}(u) = \delta_i \left[(u + V_{i,\text{ion}})\sigma_{i,\text{ion}}(u + V_{i,\text{ion}})f(u + V_{i,\text{ion}}) - u\sigma_{i,\text{ion}}(u)f(u) + \frac{1}{\gamma} \frac{\langle \nu_{i,\text{ion}} \rangle}{N} \delta(u) \right] , \quad (9)$$

corresponding to the inelastic part of (7a) plus an extra term that describes the introduction of new electrons at zero-energy;

- **equalSharing**

$$S_{i,\text{ion}}(u) = \delta_i [4(2u + V_{i,\text{ion}})\sigma_{i,\text{ion}}(2u + V_{i,\text{ion}})f(2u + V_{i,\text{ion}}) - u\sigma_{i,\text{ion}}(u)f(u)] , \quad (10)$$

where the first term of $S_{i,\text{ion}}(u)du$ describes the entrance of both the secondary and the scattered electrons at a total energy between $2u$ and $2u + 2du$, produced by a primary electron with energy $u' = 2u + V_{i,\text{ion}}$, whereas the second term describes the exit of primary electrons with energy u after ionizing collisions.

2 Discretization of the electron Boltzmann equation

2.1 The non-uniform energy grid

The EBE is discretized in a non-uniform energy grid defined in the energy interval $[0, u_{\max}]$, with *cells*

$$u_n^{\text{cell}} \equiv u_n = \sum_{p=1}^n \Delta u_p - \Delta u_n/2, \quad n = 1, 2, \dots, \mathcal{N} \quad (11)$$

of variable step-size Δu_n , each cell n being limited by *nodes* n and $n + 1$, each node n having energy

$$u_n^{\text{node}} \equiv u_{n+1}^{\text{node}} - \Delta u_n = u_n - \Delta u_n/2, \quad n = 1, 2, \dots, \mathcal{N} \quad (12a)$$

$$u_{\mathcal{N}+1}^{\text{node}} = \sum_{p=1}^{\mathcal{N}} \Delta u_p \equiv u_{\max} \quad (12b)$$

In general, the relation between each grid-cell/node and its neighbors can be arbitrarily defined.

Figure 1 illustrates graphically a non-uniform energy grid with cells and nodes defined according to the notation of (11)-(12b).

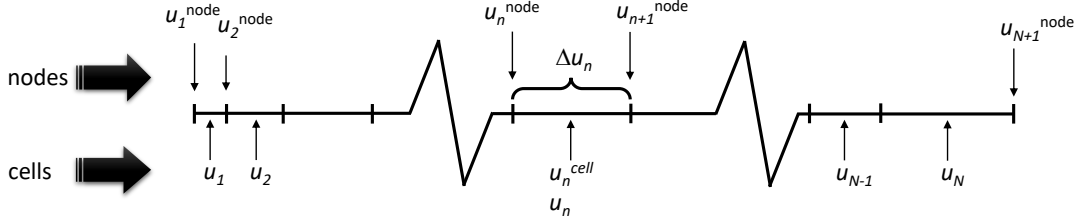


Figure 1: Graphic representation of a non-uniform energy grid with cell and nodes defined according to equations (11)-(12a).

Note that for an uniform energy grid with constant step-size Δu , equations (11)-(12b) yield

$$u_n^{\text{cell}} \equiv u_n = (n - 1/2)\Delta u, \quad n = 1, 2, \dots, \mathcal{N} \quad (13a)$$

$$u_n^{\text{node}} \equiv u_{n-1/2} = (n - 1)\Delta u, \quad n = 1, 2, \dots, \mathcal{N} + 1, \quad (13b)$$

and the graphic representation is modified as illustrated in figure 2.

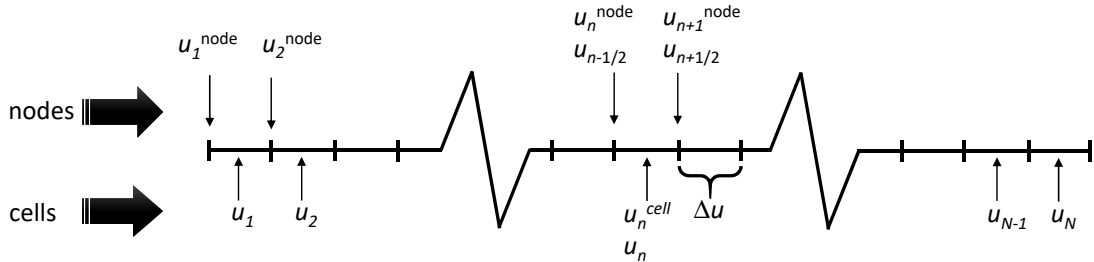


Figure 2: Graphic representation of an uniform energy grid, with constant step-size Δu , as defined by equations (13a)-(13b).

2.1.1 Defining the non-uniform energy grid

As mentioned, the location of the grid-cells/nodes of the non-uniform energy grid can be arbitrarily defined. A particular (convenient) definition can be done

- by setting the maximum energy $u_{\max}$, the number of cells $\mathcal{N}$ and the minimum energy-step $\Delta u_1 \equiv h$ ²;
- by considering consecutive energy steps that grow according to a geometric progression of ratio a

$$\Delta u_{n+1} = a \Delta u_n = a^n h, \quad a > 1, \quad (14a)$$

and therefore

$$\begin{aligned} u_1^{\text{node}} &= 0 \\ u_2^{\text{node}} &= \Delta u_1 \\ u_3^{\text{node}} &= u_2^{\text{node}} + \Delta u_2 = \Delta u_1(1 + a) \\ u_4^{\text{node}} &= u_3^{\text{node}} + \Delta u_3 = \Delta u_1(1 + a + a^2) \\ &\dots \\ u_n^{\text{node}} &= \Delta u_1(1 + a + a^2 + \dots + a^{n-2}) \\ &\dots \\ u_{\mathcal{N}}^{\text{node}} &= \Delta u_1(1 + a + a^2 + \dots + a^{\mathcal{N}-2}) \end{aligned} \quad (14b)$$

Note that, for $a = 1$ equations (14a)-(14b) retrieve the definition of an uniform energy grid with constant step-size h .

The value of the ratio a can be determined from equation (14b) using various methods.

- solving the transcendental equation resulting from the sum of the geometric progression (valid for $a \neq 1$)

$$\begin{aligned} u_{\max} &= h(1 + a + a^2 + \dots + a^{\mathcal{N}-2}) = h \frac{1 - a^{\mathcal{N}-1}}{1 - a} \\ \Rightarrow a &= \exp \left(\frac{\ln \left[\frac{u_{\max}}{h} (a - 1) + 1 \right]}{\mathcal{N} - 1} \right). \end{aligned} \quad (15a)$$

Equation (15a) must be solved iteratively, for example applying a Newton-Raphson algorithm to function

$$F(a) = \exp \left(\frac{\ln \left[\frac{u_{\max}}{h} (a - 1) + 1 \right]}{\mathcal{N} - 1} \right) - a \quad (15b)$$

which yields

$$\begin{aligned} F(a + da) &\simeq F(a) + F'(a)da = 0 \\ \Rightarrow da &= -\frac{F(a)}{F'(a)} = -\frac{\exp \left(\frac{\ln \left[\frac{u_{\max}}{h} (a - 1) + 1 \right]}{\mathcal{N} - 1} \right) - a}{\exp \left(\frac{\ln \left[\frac{u_{\max}}{h} (a - 1) + 1 \right]}{\mathcal{N} - 1} \right) \frac{u_{\max}}{[u_{\max}(a - 1) + h](\mathcal{N} - 1)}} \\ \Rightarrow da &= -\frac{[u_{\max}(a - 1) + h](\mathcal{N} - 1)}{u_{\max}} \left[1 - a \exp \left(-\frac{\ln \left[\frac{u_{\max}}{h} (a - 1) + 1 \right]}{\mathcal{N} - 1} \right) \right]. \end{aligned} \quad (15c)$$

- applying a Newton-Raphson algorithm to equation (14b)

$$\begin{aligned} F(a) &= h(1 + a + a^2 + \dots + a^{\mathcal{N}-2}) - u_{\max} \\ F(a + da) &\simeq F(a) + F'(a)da = 0 \\ \Rightarrow da &= -\frac{F(a)}{F'(a)} = -\frac{h(1 + a + a^2 + \dots + a^{\mathcal{N}-2}) - u_{\max}}{h(1 + 2a + 3a^2 + \dots + (\mathcal{N} - 2)a^{\mathcal{N}-3})}. \end{aligned} \quad (16)$$

²The maximum energy $u_{\max}$ can be defined by prescribing a number of decades in the fall of the EEDF, between f_1 and $f_{\mathcal{N}}$, as to ensure an automatic adjustment of the energy grid. Typical values for the decade-fall are within 10 to 25, considering not only the use of double precision, but also the order of magnitude of the various quantities (kinetic energy, cross sections) multiplying the EEDF in the calculations.

2.1.2 Calculating functions at grid-cells and grid-nodes

Hereafter, the functions defined at *grid-cells* n are labeled with sub-index n , and the functions defined at *grid-nodes* n are labeled with sub-index n and super-index node, for example

- for the EEDF, f , defined at grid-cells

$$\begin{aligned} f_n &\equiv f(u_n) \\ f_{n+1} &\equiv f(u_{n+1}) \quad ; \end{aligned}$$

- for the electron scattering cross section with mechanism y , σ_y , defined at grid-nodes

$$\begin{aligned} \sigma_{y_n}^{\text{node}} &\equiv \sigma_y(u_n^{\text{node}}) \\ \sigma_{y_{n+1}}^{\text{node}} &\equiv \sigma_y(u_{n+1}^{\text{node}}) \quad . \end{aligned}$$

The **linear interpolation** of physical quantities in this grid is as follows.

- **from grid-cells to grid-nodes**, taking the EEDF as example.

The EEDF at node $n + 1$, $f(u_{n+1}^{\text{node}}) \equiv f_{n+1}^{\text{node}}$, is assumed to satisfy

$$\frac{f_{n+1} - f_{n+1}^{\text{node}}}{\Delta u_{n+1}/2} = \frac{f_{n+1}^{\text{node}} - f_n}{\Delta u_n/2} \quad (17a)$$

and therefore

$$f_{n+1}^{\text{node}} = \left[\frac{f_{n+1}}{\Delta u_{n+1}} + \frac{f_n}{\Delta u_n} \right] \left[\frac{\Delta u_{n+1} \Delta u_n}{\Delta u_{n+1} + \Delta u_n} \right] = w_n f_{n+1} + (1 - w_n) f_n \quad (17b)$$

with the weight at grid-cell n given by

$$w_n \equiv \frac{\Delta u_n}{\Delta u_{n+1} + \Delta u_n} \quad . \quad (17c)$$

Note that for a uniform energy grid with constant step-size Δu , equations (17b)-(17c) yield $w_n = 1/2$ and $f_{n+1}^{\text{node}} = (f_{n+1} + f_n)/2$.

- **from grid-nodes to grid-cells**, taking the electron scattering cross section σ_y as example.

In this case, the linear interpolation gives straightforwardly

$$\sigma_y(u_n) = \frac{1}{2} \left(\sigma_{y_{n+1}}^{\text{node}} + \sigma_{y_n}^{\text{node}} \right) \quad . \quad (18)$$

The **difference between the energies of two adjacent grid-cells** is given by (see (17c))

$$u_{n+1} - u_n = \frac{1}{2} (\Delta u_{n+1} + \Delta u_n) = \frac{\Delta u_n}{2w_n} \quad . \quad (19)$$

2.1.3 Calculating the cell-number of a given energy

To discretize the discrete operators of the EBE it is relevant to calculate the cell-number X where the electron energy has a certain value u_X .

For $u_n^{\text{node}} < u_X < u_{n+1}^{\text{node}}$ and assuming a non-uniform energy grid with consecutive energy steps following a geometric progression, the cell number is defined as $X = n$ (see figure 1), therefore satisfying (see equations (14b) and (15a))

$$u_X = h \frac{a^{X-1} - 1}{a - 1} \implies X = \text{INT} \left\{ \frac{\ln \left[\frac{u_X}{h} (a - 1) + 1 \right]}{\ln a} + 1 \right\} \quad . \quad (20)$$

2.2 The discretized equations

The expressions below are to be checked / compared with those in LMarques code.
Note that LMarques discretizes the operators multiplied by the energy interval Δu_n .

Equations (2a)-(4b) are written for each *grid-cell*, where the isotropic distribution function f is defined, with the corresponding anisotropic function f^1 , upfluxes G_x ($x = E, \text{el}, \text{CAR}, ee$) and cross sections σ_y ($y = k, c; k_{(i,j)}$) being set on the *grid-nodes*³. The cross sections are interpolated linearly from the data provided by the user, setting to zero all the values up to the corresponding energy threshold V_{ij} , i.e. $\sigma_{y_n} = 0$ for n such that $u_n^{\text{node}} \leq V_{i,j}$ ⁴. The isotropic equations (2a)-(3a) are discretized with all the terms on the right-hand side; the isotropic equation (4a) is discretized with the time-dependent term on the left-hand side and all other terms on the right-hand side.

The EBE is discretised adopting a centered finite differences scheme. The result is a system of algebraic equations that can be written under matrix form, with tridiagonal elements coming from the left-hand sides of (2a) and (3a), and some sparse off-tridiagonal elements due to the electron-neutral collisional operator in the right-hand sides.

The emphasis of the discretization is with the definition of the first-anisotropy $f^1(u)$ and the upflux functions $G(u)$ on the grid-nodes, i.e. on the general boundaries $(u_n^{\text{node}}, u_{n+1}^{\text{node}})$ of a n -cell centered at energy u_n . The discretization method can be described as a finite-volume-like procedure, by applying Gauss' theorem to an integral over a one-dimensional energy volume around u_n as follows.

For the first-anisotropy function (2b)-(3b) one can write

$$\begin{aligned}
(f^1)_n^{\text{node}} &= \left[b_1(u)f(u) + b_2(u)\frac{df(u)}{du} \right]_n^{\text{node}} \\
&= \frac{1}{\Delta u_n/2 + \Delta u_{n-1}/2} \left[\int_{u_{n-1}}^{u_n} b_1(u)f(u)du + \int_{u_{n-1}}^{u_n} b_2(u)\frac{df(u)}{du}du \right] \\
&= \frac{2}{\Delta u_n + \Delta u_{n-1}} \left[\int_{u_{n-1}}^{u_n^{\text{node}}} b_1(u)f(u)du + \int_{u_n^{\text{node}}}^{u_n} b_1(u)f(u)du \right] + \\
&\quad \frac{2}{\Delta u_n + \Delta u_{n-1}} \left[\int_{u_{n-1}}^{u_n^{\text{node}}} b_2(u)\frac{df(u)}{du}du + \int_{u_n^{\text{node}}}^{u_n} b_2(u)\frac{df(u)}{du}du \right] \\
&\simeq \frac{2(b_1)_n^{\text{node}}}{\Delta u_n + \Delta u_{n-1}} \left[f_n^{\text{node}} \frac{\Delta u_{n-1}}{2} + f_n^{\text{node}} \frac{\Delta u_n}{2} \right] + \\
&\quad \frac{2(b_2)_n^{\text{node}}}{\Delta u_n + \Delta u_{n-1}} \left[\frac{(f_n^{\text{node}} - f_{n-1}) \Delta u_{n-1}/2}{\Delta u_{n-1}/2} + \frac{(f_n - f_n^{\text{node}}) \Delta u_n/2}{\Delta u_n/2} + \right] \\
&= (b_1)_n^{\text{node}} [w_{n-1}f_n + (1 - w_{n-1})f_{n-1}] + (b_2)_n^{\text{node}} \frac{f_n - f_n^{\text{node}}}{\Delta u_n/2} \quad [\text{see equations (17a)-(17b)}] \\
&= (b_1)_n^{\text{node}} [w_{n-1}f_n + (1 - w_{n-1})f_{n-1}] + (b_2)_n^{\text{node}} \frac{f_n - w_{n-1}f_n - (1 - w_{n-1})f_{n-1}}{\Delta u_n/2} \\
&= (b_1)_n^{\text{node}} [w_{n-1}f_n + (1 - w_{n-1})f_{n-1}] + (b_2)_n^{\text{node}} \frac{(1 - w_{n-1})(f_n - f_{n-1})}{\Delta u_n/2} \\
&= (b_1)_n^{\text{node}} [w_{n-1}f_n + (1 - w_{n-1})f_{n-1}] + (b_2)_n^{\text{node}} \frac{2w_{n-1}}{\Delta u_{n-1}}(f_n - f_{n-1}) \quad [\text{see equation (17c)}] \\
&= (b_1)_n^{\text{node}} f_n^{\text{node}} + (b_2)_n^{\text{node}} \frac{2}{\Delta u_n + \Delta u_{n-1}}(f_n - f_{n-1}) \quad [\text{see equation (17c)}] \quad , \quad (21)
\end{aligned}$$

where $b_1(u)$ and $b_2(u)$ are generic functions of u .

The discretization of the upflux terms in (2a)-(3a) embodies the discretization of the first-anisotropy function,

³Note that the anisotropic function f^1 is defined here on the *grid-nodes* (not the *grid-cells*), since the upflux functions G are also defined on the grid nodes and G_E is proportional to f^1 . This new framework differs from that adopted in LoKI-B v2.2.0, and ensures a consistent writing of the power absorbed from the electric field in terms of the electron mobility and free-diffusion coefficient.

⁴For an uniform grid with constant step-size Δu , this criterion writes $(n-1)\Delta u \leq V_{i,j}$ or $n \leq \text{INT}(V_{i,j}/\Delta u) + 1$. The last expression corrects that published in previous versions of the LoKI user manual.

which must be consistently defined at the grid nodes according to (21). In general, one can write

$$-\frac{1}{N} \frac{1}{\gamma} \frac{dG_x(u)}{du} \Big|_{n; x=E, \text{el}, \text{CAR}} = -\frac{1}{N} \frac{1}{\gamma} \frac{1}{\Delta u_n} \int_{u_n^{\text{node}}}^{u_{n+1}^{\text{node}}} \frac{dG_x(u)}{du} du = -\frac{1}{N} \frac{1}{\gamma} \frac{G_{x_{n+1}}^{\text{node}} - G_{x_n}^{\text{node}}}{\Delta u_n}, \quad (22a)$$

and therefore ⁵

$$\begin{aligned} -\frac{1}{N} \frac{1}{\gamma} \frac{dG_x(u)}{du} \Big|_{n; x=E, \text{el}, \text{CAR}} &= \left\{ \frac{d}{du} \left[g_x(u) \left(d_x f(u) + c_x \frac{df(u)}{du} \right) \right] \right\}_n \\ &\simeq \frac{g_{x_{n+1}}^{\text{node}}}{\Delta u_n} \frac{2}{\Delta u_n + \Delta u_{n+1}} \left(d_x \int_{u_n}^{u_{n+1}} f(u) du + c_x \int_{u_n}^{u_{n+1}} \frac{df(u)}{du} du \right) \\ &\quad - \frac{g_{x_n}^{\text{node}}}{\Delta u_n} \frac{2}{\Delta u_{n-1} + \Delta u_n} \left(d_x \int_{u_{n-1}}^{u_n} f(u) du + c_x \int_{u_{n-1}}^{u_n} \frac{df(u)}{du} du \right) \\ &\simeq \frac{g_{x_{n+1}}^{\text{node}}}{\Delta u_n} \left\{ d_x [(1 - w_n) f_n + w_n f_{n+1}] + c_x \frac{2w_n}{\Delta u_n} (f_{n+1} - f_n) \right\} \\ &\quad - \frac{g_{x_n}^{\text{node}}}{\Delta u_n} \left\{ d_x [(1 - w_{n-1}) f_{n-1} + w_{n-1} f_n] + c_x \frac{2w_{n-1}}{\Delta u_{n-1}} (f_n - f_{n-1}) \right\} \\ &= \left\{ \frac{g_{x_{n+1}}^{\text{node}}}{\Delta u_n} \left[c_x \frac{2w_{n-1}}{\Delta u_{n-1}} - d_x (1 - w_{n-1}) \right] \right\} f_{n-1} \\ &\quad - \left\{ \frac{g_{x_{n+1}}^{\text{node}}}{\Delta u_n} \left[c_x \frac{2w_n}{\Delta u_n} - d_x (1 - w_n) \right] + \frac{g_{x_n}^{\text{node}}}{\Delta u_n} \left[c_x \frac{2w_{n-1}}{\Delta u_{n-1}} + d_x w_{n-1} \right] \right\} f_n \\ &\quad + \left\{ \frac{g_{x_{n+1}}^{\text{node}}}{\Delta u_n} \left[c_x \frac{2w_n}{\Delta u_n} + d_x w_n \right] \right\} f_{n+1} \quad , \end{aligned} \quad (22b)$$

where c_x and d_x are constants, and $g_x(u)$ can be obtained straightforwardly from (6a)-(6c).

Note that equation (21) is used also in the discretization of the spatial growth term in (3a).

In addition to the terms proportional to f^1 and $dG_{x=E, \text{el}, \text{CAR}}/du$, equations (2a)-(4b) are further composed by terms proportional to f and dG_{ee}/du whose discretised forms can be generally summarised as follows (the latter receiving a treatment similar to equations (22a)-(22b))

$$[a(u)f(u)]_n = \frac{1}{\Delta u_n} \int_{u_n^{\text{node}}}^{u_{n+1}^{\text{node}}} a(u)f(u) du = a_n f_n \quad (23a)$$

$$[a'(u)f(u \pm V_{i,j})]_n = a'_n f_{m_n} \quad [\text{see item **Discrete collision operator** below}] \quad (23b)$$

$$-\frac{1}{N} \frac{1}{\gamma} \frac{dG_{ee}(u)}{du} \Big|_n = \left\{ \frac{d}{du} \left[g_{ee}(u) \left(I(u)f(u) + J(u) \frac{df(u)}{du} \right) \right] \right\}_n \quad , \quad (23c)$$

where $a(u)$ and $a'(u)$ are generic functions of u ; and $g_{ee}(u)$, $I(u)$ and $J(u)$ can be obtained straightforwardly from (6d).

The detailed discretized form of each term is given below (the titles marked with $\star$, signal the terms with modified expressions for a non-uniform energy grid).

Two important notes.

1. The following expressions are written for a generic non-uniform energy grid, *i.e.* they apply to grids that don't satisfy the definition of subsection 2.1.3.
2. The following expressions yield the usual discretization in a uniform energy grid with constant step-size Δu , when taking $\Delta u_n = \Delta u_m = \dots = \Delta u$ and $w_n = w_m = \dots = 1/2$.

⁵In the LoKI-B v2.2.0 user manual, a factor $1/\Delta u$ is missing in the 2nd and 3rd expressions of equation (22b).

- Time-dependent term

$$\frac{1}{N} \frac{1}{\gamma} \sqrt{u} \frac{\partial f(u, t)}{\partial t} \Big|_n = \frac{1}{N} \frac{1}{\gamma} u_n^{1/2} \frac{\partial f_n(t)}{\partial t} . \quad (24)$$

- Temporal growth term

$$-\frac{1}{\gamma \sqrt{u}} \frac{\langle \nu_{\text{eff}} \rangle}{N} u f(u) \Big|_n = -\frac{1}{\gamma} \frac{\langle \nu_{\text{eff}} \rangle}{N} u_n^{1/2} f_n . \quad (25)$$

- Spatial growth term^{*} ⁶

$$\begin{aligned} -\frac{1}{3} \frac{\alpha_{\text{eff}}}{N} u f^1(u) \Big|_n &= \frac{1}{3} \frac{\alpha_{\text{eff}}}{N} u \left[\frac{(\alpha_{\text{eff}}/N)}{\Omega_{\text{SST}}(u)} f(u) + \frac{(E/N)}{\Omega_{\text{SST}}(u)} \frac{df(u)}{du} \right] \Big|_n = -\frac{1}{3} \frac{\alpha_{\text{eff}}}{N} \frac{u_n^{\text{node}} f_{n+1}^{1, \text{node}} + u_n^{\text{node}} f_n^{1, \text{node}}}{2} \\ &= \frac{1}{3} \frac{\alpha_{\text{eff}}}{N} \left\{ \frac{u_{n+1}^{\text{node}} \alpha_{\text{eff}}/N}{2\Omega_{\text{SST}_{n+1}}^{\text{node}}} [w_n f_{n+1} + (1 - w_n) f_n] + \frac{u_n^{\text{node}} \alpha_{\text{eff}}/N}{2\Omega_{\text{SST}_n}^{\text{node}}} [w_{n-1} f_n + (1 - w_{n-1}) f_{n-1}] \right. \\ &\quad \left. + \frac{u_{n+1}^{\text{node}} E/N}{2\Omega_{\text{SST}_{n+1}}^{\text{node}}} \frac{2(f_{n+1} - f_n)}{\Delta u_{n+1} + \Delta u_n} + \frac{u_n^{\text{node}} E/N}{2\Omega_{\text{SST}_n}^{\text{node}}} \frac{2(f_n - f_{n-1})}{\Delta u_n + \Delta u_{n-1}} \right\} . \end{aligned} \quad (26)$$

- Electric field operator^{*}

$$g_E(u) = \frac{u}{3} \frac{E}{N} \times \begin{cases} \frac{(E/N)}{\Omega_{\text{PT}}(u)} , & \text{for a PT situation} \\ \frac{(\alpha_{\text{eff}}/N)}{\Omega_{\text{SST}}(u)} , & \text{for a SST situation} \end{cases} \quad (27a)$$

$$c_E = \begin{cases} 1 , & \text{for a PT situation} \\ \frac{(E/N)}{(\alpha_{\text{eff}}/N)} , & \text{for a SST situation} \end{cases} \quad (27b)$$

$$d_E = \begin{cases} 0 , & \text{for a PT situation} \\ 1 , & \text{for a SST situation} \end{cases} \quad (27c)$$

For a PT situation

$$-\frac{1}{N} \frac{1}{\gamma} \frac{dG_E(u)}{du} \Big|_n = \left[\frac{g_{E_n}^{\text{node}} 2w_{n-1}}{\Delta u_n \Delta u_{n-1}} \right] f_{n-1} - \left[\frac{g_{E_{n+1}}^{\text{node}} 2w_n}{(\Delta u_n)^2} + \frac{g_{E_n}^{\text{node}} 2w_{n-1}}{\Delta u_n \Delta u_{n-1}} \right] f_n + \left[\frac{g_{E_{n+1}}^{\text{node}} 2w_n}{(\Delta u_n)^2} \right] f_{n+1} . \quad (28)$$

For an SST situation

$$\begin{aligned} -\frac{1}{N} \frac{1}{\gamma} \frac{dG_E(u)}{du} \Big|_n &= \left\{ \frac{g_{E_n}^{\text{node}}}{\Delta u_n} \left[\frac{E/N}{(\alpha_{\text{eff}}/N)} \frac{2w_{n-1}}{\Delta u_{n-1}} - (1 - w_{n-1}) \right] \right\} f_{n-1} \\ &\quad - \left\{ \frac{g_{E_{n+1}}^{\text{node}}}{\Delta u_n} \left[\frac{E/N}{(\alpha_{\text{eff}}/N)} \frac{2w_n}{\Delta u_n} - (1 - w_n) \right] + \frac{g_{E_n}^{\text{node}}}{\Delta u_n} \left[\frac{E/N}{(\alpha_{\text{eff}}/N)} \frac{2w_{n-1}}{\Delta u_{n-1}} + w_{n-1} \right] \right\} f_n \\ &\quad + \left\{ \frac{g_{E_{n+1}}^{\text{node}}}{\Delta u_n} \left[\frac{E/N}{(\alpha_{\text{eff}}/N)} \frac{2w_n}{\Delta u_n} + w_n \right] \right\} f_{n+1} \end{aligned} \quad (29)$$

- Elastic operator^{*}

$$g_{\text{el}}(u) = 2u^2 \sum_k \frac{m_e}{M_k} \chi_k \sigma_{k,c}^{\text{el}}(u) \quad (30a)$$

$$c_{\text{el}} = \frac{k_B T_g}{e} \quad (30b)$$

$$d_{\text{el}} = 1 \quad (30c)$$

⁶The expressions below adopt a discretization framework different from that of LoKI-B v2.2.0, since they define the anisotropic function f^1 on the grid-nodes (not the grid-cells). Thus, they **don't** yield the usual discretization in a uniform energy grid.

$$\begin{aligned}
-\frac{1}{N} \frac{1}{\gamma} \frac{dG_{\text{el}}(u)}{du} \Big|_n &= \left\{ \frac{g_{\text{el}_n}^{\text{node}}}{\Delta u_n} \left[\frac{k_B T_g}{e} \frac{2w_{n-1}}{\Delta u_{n-1}} - (1 - w_{n-1}) \right] \right\} f_{n-1} \\
&- \left\{ \frac{g_{\text{el}_{n+1}}^{\text{node}}}{\Delta u_n} \left[\frac{k_B T_g}{e} \frac{2w_n}{\Delta u_n} - (1 - w_n) \right] + \frac{g_{\text{el}_n}^{\text{node}}}{\Delta u_n} \left[\frac{k_B T_g}{e} \frac{2w_{n-1}}{\Delta u_{n-1}} + w_{n-1} \right] \right\} f_n \quad . \quad (31) \\
&+ \left\{ \frac{g_{\text{el}_{n+1}}^{\text{node}}}{\Delta u_n} \left[\frac{k_B T_g}{e} \frac{2w_n}{\Delta u_n} + w_n \right] \right\} f_{n+1}
\end{aligned}$$

• **CAR operator***

$$g_{\text{CAR}}(u) = 4u \sum_k B_k \chi_k \sigma_{0,k}(u) \quad (32a)$$

$$c_{\text{CAR}} = \frac{k_B T_g}{e} \quad (32b)$$

$$d_{\text{CAR}} = 1 \quad (32c)$$

$$\begin{aligned}
-\frac{1}{N} \frac{1}{\gamma} \frac{dG_{\text{CAR}}(u)}{du} \Big|_n &= \left\{ \frac{g_{\text{CAR}_n}^{\text{node}}}{\Delta u_n} \left[\frac{k_B T_g}{e} \frac{2w_{n-1}}{\Delta u_{n-1}} - (1 - w_{n-1}) \right] \right\} f_{n-1} \\
&- \left\{ \frac{g_{\text{CAR}_{n+1}}^{\text{node}}}{\Delta u_n} \left[\frac{k_B T_g}{e} \frac{2w_n}{\Delta u_n} - (1 - w_n) \right] + \frac{g_{\text{CAR}_n}^{\text{node}}}{\Delta u_n} \left[\frac{k_B T_g}{e} \frac{2w_{n-1}}{\Delta u_{n-1}} + w_{n-1} \right] \right\} f_n \quad . \\
&+ \left\{ \frac{g_{\text{CAR}_{n+1}}^{\text{node}}}{\Delta u_n} \left[\frac{k_B T_g}{e} \frac{2w_n}{\Delta u_n} + w_n \right] \right\} f_{n+1}
\end{aligned} \quad (33)$$

• **Electron-electron collisions operator***⁷

[In the case of time-dependent simulations, the electron density $n_e(t)$ and the EEDF $f(u, t)$ introduce a time-dependence into $g_{ee}(u, t)$, $I(u, t)$ and $J(u, t)$, which are written for a given t .]

$$g_{ee}(u) = \frac{2}{\gamma} \frac{\nu_{ee}(u)}{N} u^{3/2} \quad (34a)$$

$$I(u) = \int_0^u f(u') u'^{1/2} du' \quad (34b)$$

$$J(u) = \frac{2}{3} \left[\int_0^u f(u') u'^{3/2} du' + u^{3/2} \int_u^\infty f(u') du' \right] \quad (34c)$$

$$\begin{aligned}
-\frac{1}{N} \frac{1}{\gamma} \frac{dG_{ee}(u)}{du} \Big|_n &= \left\{ \frac{d}{du} \left[g_{ee}(u) \left(I(u) f(u) + J(u) \frac{df(u)}{du} \right) \right] \right\}_n \\
&= \frac{1}{\Delta u_n} [A_{n-1} f_{n-1} - (A_n + B_n) f_n + B_{n+1} f_{n+1}] \quad . \\
&= \frac{1}{\Delta u_n} \left[\sum_{m=1}^{\mathcal{N}} (a_{n-1,m} f_m) f_{n-1} - \sum_{m=1}^{\mathcal{N}} (a_{n,m} f_m + b_{n,m} f_m) f_n + \sum_{m=1}^{\mathcal{N}} (b_{n+1,m} f_m) f_{n+1} \right] \quad (35)
\end{aligned}$$

⁷The expressions below redefine the matrix elements A_n , B_n and $a_{n,m}$, $b_{n,m}$, with respect to LoKI-B v2.2.0. Here, these elements are defined with an extra multiplication by the energy step Δu_n .

$$\begin{aligned}
A_n &\equiv A_{I_n} + A_{J_n} \equiv \sum_{m=1}^{\mathcal{N}} a_{n,m} f_m \\
&= -\frac{2}{\gamma} \frac{\nu_{ee_{n+1}}^{\text{node}}}{N} (u_{n+1}^{\text{node}})^{3/2} \left[I_{n+1}^{\text{node}} (1 - w_n) - J_{n+1}^{\text{node}} \frac{2w_n}{\Delta u_n} \right]
\end{aligned} \tag{36a}$$

$$\begin{aligned}
B_n &\equiv B_{I_n} + B_{J_n} \equiv \sum_{m=1}^{\mathcal{N}} b_{n,m} f_m \\
&= \frac{2}{\gamma} \frac{\nu_{ee_n}^{\text{node}}}{N} (u_n^{\text{node}})^{3/2} \left[I_n^{\text{node}} w_{n-1} + J_n^{\text{node}} \frac{2w_{n-1}}{\Delta u_{n-1}} \right]
\end{aligned} \tag{36b}$$

$$A_{I_n} = -\frac{2}{\gamma} \frac{\nu_{ee_{n+1}}^{\text{node}}}{N} (u_{n+1}^{\text{node}})^{3/2} I_{n+1}^{\text{node}} (1 - w_n) \quad , \quad \text{etc} \tag{36c}$$

$$A_{J_n} = A_n w_n + B_{n+1} (1 - w_n) \tag{36d}$$

$$B_{J_n} = A_{n-1} w_{n-1} + B_n (1 - w_{n-1}) \tag{36e}$$

$$\frac{\nu_{ee_n}^{\text{node}}}{N} (u_n^{\text{node}})^{3/2} = \pi \left(\frac{e^2}{4\pi\epsilon_0 m_e} \right)^2 \left(\frac{m_e}{2e} \right)^{3/2} \ln \left[12\pi n_e \left(\frac{\epsilon_0 T_e}{n_e e} \right)^{3/2} \right] \frac{n_e}{N} \tag{36f}$$

$$T_e = \frac{2}{3} \sum_{m=1}^{\mathcal{N}} f_m u_m^{3/2} \Delta u_m \tag{36g}$$

$$I_{n+1}^{\text{node}} = \sum_{m=1}^n f_m u_m^{1/2} \Delta u_m \tag{36h}$$

$$J_{n+1}^{\text{node}} = \frac{2}{3} \left[\sum_{m=1}^n f_m u_m^{3/2} \Delta u_m + (u_{n+1}^{\text{node}})^{3/2} \sum_{m=n+1}^{\mathcal{N}} f_m \Delta u_m \right] \tag{36i}$$

$$a_{n,m} = -\frac{2}{\gamma} \frac{\nu_{ee_{n+1}}^{\text{node}}}{N} (u_{n+1}^{\text{node}})^{3/2} \begin{cases} (1 - w_n) u_m^{1/2} \Delta u_m - \frac{2}{3} \frac{2w_n}{\Delta u_n} u_m^{3/2} \Delta u_m \quad , \quad m \leq n \\ -\frac{2}{3} \frac{2w_n}{\Delta u_n} (u_{n+1}^{\text{node}})^{3/2} \Delta u_m \quad , \quad m > n \end{cases} \tag{36j}$$

$$b_{n,m} = \frac{2}{\gamma} \frac{\nu_{ee_n}^{\text{node}}}{N} (u_n^{\text{node}})^{3/2} \times \begin{cases} w_{n-1} u_m^{1/2} \Delta u_m + \frac{2}{3} \frac{2w_{n-1}}{\Delta u_{n-1}} u_m^{3/2} \Delta u_m \quad , \quad m \leq n-1 \\ \frac{2}{3} \frac{2w_{n-1}}{\Delta u_{n-1}} (u_n^{\text{node}})^{3/2} \Delta u_m \quad , \quad m > n-1 \end{cases} \tag{36k}$$

• Discrete collision operators*

In general, the discretization monitors the shift of electrons between high/low energy intervals with different sizes,

by **evaluating the relationship between EEDFs** defined at the *location from where an electron is removed*, and the *current location that receives the electron*.

Specifically, this mean that the discretization focuses on the *entrance operators*, monitoring the entry of electrons into an energy interval $\Delta u_n = u_{n+1}^{\text{node}} - u_n^{\text{node}}$, due to ⁸

- inelastic deexcitations with energy threshold V_{ij} , from higher energies starting at $u_n^{\text{node}} + V_{i,j} \leq u_{m_n+1}^{\text{node}}$;
- superelastic excitations with energy threshold V_{ij} , from lower energies starting at $u_n^{\text{node}} - V_{i,j} \geq u_{m_n}^{\text{node}}$;

by **ensuring the conservation of particles**, *i.e.* $\sum_n S_n^{\text{inel}} \Delta u_n = \sum_n S_n^{\text{sup}} \Delta u_n = 0$ (see section 2.4), defining the *exit operators* from the corresponding entrance components.

In the following, the labels “inel” and “sup” refer to inelastic and superelastic collisions, respectively, associated with electronic, vibrational and rotational excitation / de-excitation mechanisms.

⁸Strickly, the index m_n should be replaced by $m_{n,(i,j)}$, expliciting the link with the entry-cell n and the specific transition $i \leftrightarrow j$. For simplicity, we have chosen to omit the subindex (i,j) , since the analysis will be made mostly for one specific transition, the generalization to all transitions being obtained by summing over all (i,j) indexes.

– **Inelastic collision operator**

I have doubts about the boundary conditions adopted in LMarques' code for this operator.

The operator is separated into its entry(+) and exit(-) components, relative to the u -energy cell (see equation (7a)),

$$S_{(i,j),\text{inel}}(u) = S_{(i,j),\text{inel}}^{(+)}(u) + S_{(i,j),\text{inel}}^{(-)}(u) \quad (37a)$$

$$S_{(i,j),\text{inel}}^{(+)}(u) = \delta_i(u + V_{i,j})\sigma_{i,j}(u + V_{i,j})f(u + V_{i,j}) \quad (37b)$$

$$S_{(i,j),\text{inel}}^{(-)}(u) = -\delta_i u \sigma_{i,j}(u)f(u) = -S_{(i,j),\text{inel}}^{(+)}(u - V_{i,j}) \quad (37c)$$

which are discretized considering the *energy exchanged in each collision* and its *location within the non-uniform grid*.

As indicated above, the discretization monitors the shift of an electron due to deexcitations with threshold $V_{i,j}$, from higher energies starting at $u_n^{\text{node}} + V_{i,j} \leq u_{m_n+1}^{\text{node}}$ into a lower-energy interval $\Delta u_n = u_{n+1}^{\text{node}} - u_n^{\text{node}}$ (see figure 3).

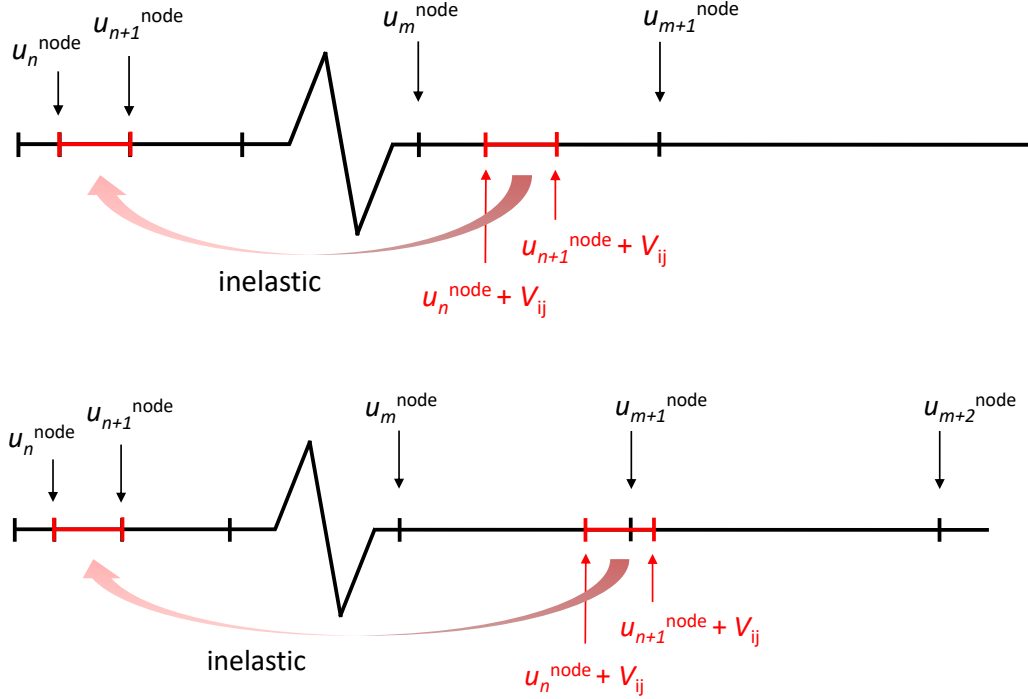


Figure 3: Graphic representation of a generic inelastic collision in a non-uniform energy grid.

Because $\Delta u_n \leq \Delta u_{m_n}$ (the equality holding in the case of an uniform energy-grid only), we separate the situations where either

- (i) Δu_n falls entirely within the larger interval Δu_{m_n} , i.e. $u_{m_n}^{\text{node}} \leq u_n^{\text{node}} + V_{i,j}$, $u_{n+1}^{\text{node}} + V_{i,j} \leq u_{m_n+1}^{\text{node}}$;
- (ii) Δu_n is settled between the adjacent intervals Δu_{m_n} and Δu_{m_n+1} , i.e. $u_{m_n}^{\text{node}} < u_n^{\text{node}} + V_{i,j} < u_{m_n+1}^{\text{node}} < u_{n+1}^{\text{node}} + V_{i,j} < u_{m_n+2}^{\text{node}}$.

Therefore, the operator is discretized as follows with focus on its entrance component.

$$S_n^{\text{inel}} = \sum_{i,j>i} S_{(i,j)_n}^{\text{inel}} = \sum_{i,j>i} \left[S_{(i,j)_n}^{\text{inel},(+)} + S_{(i,j)_n}^{\text{inel},(-)} \right] \quad (38a)$$

$$\begin{aligned}
S_{(i,j)_n}^{\text{inel},(+)} \Delta u_n &= \begin{cases} \int_{u_n^{\text{node}}}^{u_{n+1}^{\text{node}}} S_{(i,j)}^{\text{inel},(+)}(u) du \\ \text{if } u_{m_n}^{\text{node}} \leq u_n^{\text{node}} + V_{i,j}, u_{n+1}^{\text{node}} + V_{i,j} \leq u_{m_n+1}^{\text{node}} \end{cases} \\
&= \begin{cases} \int_{u_n^{\text{node}}}^{u_{m_n+1}^{\text{node}} - V_{i,j}} S_{(i,j)}^{\text{inel},(+)}(u) du + \int_{u_{m_n+1}^{\text{node}} - V_{i,j}}^{u_{n+1}^{\text{node}}} S_{(i,j)}^{\text{inel},(+)}(u) du \\ \text{if } u_{m_n}^{\text{node}} < u_n^{\text{node}} + V_{i,j} < u_{m_n+1}^{\text{node}} < u_{n+1}^{\text{node}} + V_{i,j} < u_{m_n+2}^{\text{node}} \end{cases} \\
&= \begin{cases} \int_{u_n^{\text{node}}}^{u_{n+1}^{\text{node}}} \delta_i \sigma_{i,j}(u + V_{i,j}) f(u + V_{i,j})(u + V_{i,j}) du \\ \text{if } u_{m_n}^{\text{node}} \leq u_n^{\text{node}} + V_{i,j}, u_{n+1}^{\text{node}} + V_{i,j} \leq u_{m_n+1}^{\text{node}} \end{cases} \\
&= \begin{cases} \int_{u_n^{\text{node}}}^{u_{m_n+1}^{\text{node}} - V_{i,j}} \delta_i \sigma_{i,j}(u + V_{i,j}) f(u + V_{i,j})(u + V_{i,j}) du + \\ \int_{u_{m_n+1}^{\text{node}} - V_{i,j}}^{u_{n+1}^{\text{node}}} \delta_i \sigma_{i,j}(u + V_{i,j}) f(u + V_{i,j})(u + V_{i,j}) du \\ \text{if } u_{m_n}^{\text{node}} < u_n^{\text{node}} + V_{i,j} < u_{m_n+1}^{\text{node}} < u_{n+1}^{\text{node}} + V_{i,j} < u_{m_n+2}^{\text{node}} \end{cases} \\
&= \begin{cases} \delta_i \sigma_{(i,j)_{m_n}} f_{m_n} \left[\frac{1}{2} (u_n^{\text{node}} + u_{n+1}^{\text{node}}) + V_{i,j} \right] (u_{n+1}^{\text{node}} - u_n^{\text{node}}) \\ \text{if } u_{m_n}^{\text{node}} \leq u_n^{\text{node}} + V_{i,j}, u_{n+1}^{\text{node}} + V_{i,j} \leq u_{m_n+1}^{\text{node}} \end{cases} \\
&= \begin{cases} \delta_i \left\{ \sigma_{(i,j)_{m_n}} f_{m_n} \frac{1}{2} (u_n^{\text{node}} + V_{i,j} + u_{m_n+1}^{\text{node}}) [(u_{m_n+1}^{\text{node}} - V_{i,j}) - u_n^{\text{node}}] + \right. \\ \left. \sigma_{(i,j)_{m_n+1}} f_{m_n+1} \frac{1}{2} (u_{n+1}^{\text{node}} + V_{i,j} + u_{m_n+1}^{\text{node}}) [u_{n+1}^{\text{node}} - (u_{m_n+1}^{\text{node}} - V_{i,j})] \right\} \\ \text{if } u_{m_n}^{\text{node}} < u_n^{\text{node}} + V_{i,j} < u_{m_n+1}^{\text{node}} < u_{n+1}^{\text{node}} + V_{i,j} < u_{m_n+2}^{\text{node}} \end{cases} \\
&= \begin{cases} \delta_i \sigma_{(i,j)_{m_n}} f_{m_n} \frac{1}{2} [(u_n^{\text{node}} + V_{i,j}) + (u_{n+1}^{\text{node}} + V_{i,j})] [(u_{n+1}^{\text{node}} + V_{i,j}) - (u_n^{\text{node}} + V_{i,j})] \\ \text{if } u_{m_n}^{\text{node}} \leq u_n^{\text{node}} + V_{i,j}, u_{n+1}^{\text{node}} + V_{i,j} \leq u_{m_n+1}^{\text{node}} \end{cases} \\
&= \begin{cases} \delta_i \left\{ \sigma_{(i,j)_{m_n}} f_{m_n} \frac{1}{2} (u_n^{\text{node}} + V_{i,j} + u_{m_n+1}^{\text{node}}) [u_{m_n+1}^{\text{node}} - (u_n^{\text{node}} + V_{i,j})] + \right. \\ \left. \sigma_{(i,j)_{m_n+1}} f_{m_n+1} \frac{1}{2} (u_{n+1}^{\text{node}} + V_{i,j} + u_{m_n+1}^{\text{node}}) [(u_{n+1}^{\text{node}} + V_{i,j}) - u_{m_n+1}^{\text{node}}] \right\} \\ \text{if } u_{m_n}^{\text{node}} < u_n^{\text{node}} + V_{i,j} < u_{m_n+1}^{\text{node}} < u_{n+1}^{\text{node}} + V_{i,j} < u_{m_n+2}^{\text{node}} \end{cases} \\
&= \begin{cases} \delta_i \sigma_{(i,j)_{m_n}} f_{m_n} \frac{1}{2} \left[(u_{c_{n+1}}^{i,j})^2 - (u_{c_n}^{i,j})^2 \right] \\ \text{if } u_{m_n}^{\text{node}} \leq u_{c_n}^{i,j}, u_{c_{n+1}}^{i,j} \leq u_{m_n+1}^{\text{node}} \end{cases} \\
&= \begin{cases} \delta_i \left\{ \sigma_{(i,j)_{m_n}} f_{m_n} \frac{1}{2} \left[(u_{m_n+1}^{\text{node}})^2 - (u_{c_n}^{i,j})^2 \right] + \right. \\ \left. \sigma_{(i,j)_{m_n+1}} f_{m_n+1} \frac{1}{2} \left[(u_{c_{n+1}}^{i,j})^2 - (u_{m_n+1}^{\text{node}})^2 \right] \right\} \\ \text{if } u_{m_n}^{\text{node}} < u_{c_n}^{i,j} < u_{m_n+1}^{\text{node}} < u_{c_{n+1}}^{i,j} < u_{m_n+2}^{\text{node}} \end{cases}, \tag{38b}
\end{aligned}$$

with

$$u_{c_n}^{i,j} \equiv u_n^{\text{node}} + V_{i,j} \quad . \tag{38c}$$

Moreover,

$$\begin{aligned}
S_{(i,j)_n}^{\text{inel},(+)} \Delta u_n &= \int_{u_n^{\text{node}}}^{u_{n+1}^{\text{node}}} S_{(i,j)}^{\text{inel},(+)}(u) du \\
&= - \int_{u_n^{\text{node}}}^{u_{n+1}^{\text{node}}} S_{(i,j)}^{\text{inel},(-)}(u + V_{i,j}) du \\
&= - \int_{u_n^{\text{node}} + V_{i,j}}^{u_{n+1}^{\text{node}} + V_{i,j}} S_{(i,j)}^{\text{inel},(-)}(u) du \\
&\equiv - \sum_{k=m_n}^{m_n+1} S_{(i,j)_k}^{\text{inel},(-)} \Delta u_k \quad , \tag{38d}
\end{aligned}$$

since electrons coming out from both cells m_n and $m_n + 1$ will enter the single cell n ⁹.

Finally

$$S_{(i,j)_n}^{\text{inel},(+)} \equiv \sum_{k=m_n}^{m_n+1} C_{n,k}^{(i,j),\text{inel}} f_k = \begin{cases} \delta_i \frac{(u_{c_{n+1}}^{i,j})^2 - (u_{c_n}^{i,j})^2}{2\Delta u_n} \sigma_{(i,j)_{m_n}} f_{m_n} \\ \text{if } u_{m_n}^{\text{node}} \leq u_{c_n}^{i,j}, u_{c_{n+1}}^{i,j} \leq u_{m_n+1}^{\text{node}} \\ \delta_i \frac{(u_{m_n+1}^{\text{node}})^2 - (u_{c_n}^{i,j})^2}{2\Delta u_n} \sigma_{(i,j)_{m_n}} f_{m_n} \\ - \delta_i \frac{(u_{c_{n+1}}^{i,j})^2 - (u_{m_n+1}^{\text{node}})^2}{2\Delta u_n} \sigma_{(i,j)_{m_n+1}} f_{m_n+1} \\ \text{if } u_{m_n}^{\text{node}} < u_{c_n}^{i,j} < u_{m_n+1}^{\text{node}} < u_{c_{n+1}}^{i,j} < u_{m_n+2}^{\text{node}} \end{cases} \tag{39a}$$

$$S_{(i,j)_k}^{\text{inel},(-)} \equiv C_{k,k}^{(i,j),\text{inel}} f_k = -C_{n,k}^{(i,j),\text{inel}} \frac{\Delta u_n}{\Delta u_k} f_k \quad , \text{for } k = m_n, m_n + 1 \tag{39b}$$

$$\begin{aligned}
\Rightarrow \sum_n S_{(i,j)_n}^{\text{inel},(+)} \Delta u_n &= \sum_n \sum_{k=m_n}^{m_n+1} C_{n,k}^{(i,j),\text{inel}} f_k \Delta u_n \\
&= - \sum_n \sum_{k=m_n}^{m_n+1} C_{k,k}^{(i,j),\text{inel}} f_k \Delta u_k = - \sum_n \sum_{k=m_n}^{m_n+1} S_{(i,j)_k}^{\text{inel},(-)} \Delta u_k \quad , \tag{39c}
\end{aligned}$$

for $n, m_n = 1, 2, \dots, \mathcal{N}$ with $C_{n,\mathcal{N}+1}^{(i,j),\text{inel}} = 0$.

The previous expressions allow retrieving the usual equation of the discretized inelastic operator in a uniform energy grid, after taking into account the following differences in the methodology of discretization.

- * The criteria for defining the upper energy-interval compares the energies defined in grid-cells instead of grid-nodes.

Therefore, the condition

$$u_{m_n}^{\text{node}} \leq u_n^{\text{node}} + V_{i,j}, u_{n+1}^{\text{node}} + V_{i,j} \leq u_{m_n+1}^{\text{node}}$$

should be transformed into

$$\begin{aligned}
u_{m_n}^{\text{node}} + \frac{\Delta u}{2} &\leq u_n^{\text{node}} + \frac{\Delta u}{2} + V_{i,j}, u_{n+1}^{\text{node}} + \frac{\Delta u}{2} + V_{i,j} \leq u_{m_n+1}^{\text{node}} + \frac{\Delta u}{2} \\
u_{m_n} &\leq u_n + V_{i,j}, u_{n+1} + V_{i,j} \leq u_{m_n+1} \quad .
\end{aligned}$$

⁹Note that the definitions of $S_{(i,j)_n}^{\text{inel},(+)}$ and $S_{(i,j)_n}^{\text{inel},(-)}$ are merely symbolic. For the numerical implementation of both, the Boltzmann equation and the conservation laws, the matrix elements $C_{n,k}^{\text{inel}}$ should be used instead.

- * The excitation potentials V_{ij} are discretized with a truncation to the energy of closest grid-node using

$$l_{ij} \equiv \text{INT} \left(\frac{V_{i,j}}{\Delta u} \right) .$$

In this case, the previous condition can be written as

$$\begin{aligned} u_{m_n} &\leq u_n + l_{i,j} \Delta u, u_{n+1} + l_{i,j} \Delta u \leq u_{m_n+1} \\ u_{m_n} &\leq u_{n+l_{i,j}}, u_{n+l_{i,j}+1} \leq u_{m_n+1} , \end{aligned}$$

which allows identifying

$$\begin{aligned} m_n &= n + l_{ij} \\ u_{m_n} &= u_{n+l_{i,j}} ; u_{n+l_{i,j}+1} = u_{m_n+1} \end{aligned}$$

thus confirm that the condition holds in the case of an uniform grid.

Finally, the expression of the discretized inelastic operator in a uniform energy grid writes ¹⁰

$$\begin{aligned} S_{(i,j)_n}^{\text{inel},(+)} \Delta u_n &= -S_{(i,j)_{m_n}}^{\text{inel},(-)} \Delta u_{m_n} = \delta_i \sigma_{(i,j)_{m_n}} f_{m_n} \left[\frac{1}{2} (u_n^{\text{node}} + u_{n+1}^{\text{node}}) + V_{i,j} \right] (u_{n+1}^{\text{node}} - u_n^{\text{node}}) \\ \implies S_{(i,j)_n}^{\text{inel},(+)} &= -S_{(i,j)_{n+l_{ij}}}^{\text{inel},(-)} = \delta_i [u_n + V_{i,j}] \sigma_{(i,j)_{n+l_{ij}}} f_{n+l_{ij}} \\ \implies S_{(i,j)_n}^{\text{inel},(+)} &= -S_{(i,j)_{n+l_{ij}}}^{\text{inel},(-)} = \delta_i u_{n+l_{ij}} \sigma_{(i,j)_{n+l_{ij}}} f_{n+l_{ij}} \\ \implies S_{(i,j)_n}^{\text{inel}} &= S_{(i,j)_n}^{\text{inel},(+)} + S_{(i,j)_n}^{\text{inel},(-)} = \delta_i \left[u_{n+l_{ij}} \sigma_{(i,j)_{n+l_{ij}}} f_{n+l_{ij}} - u_n \sigma_{(i,j)_n} f_n \right] \end{aligned}$$

– Superelastic collision operator

Question to LMarques - in your code, is this operator written using inelastic cross sections, related with the superelastic ones via the Klein-Rosseland relation ?

Similarly to the inelastic collision operator, the corresponding superelastic writes (see equation (7a)),

$$S_{(i,j),\text{sup}}(u) = S_{(i,j),\text{sup}}^{(+)}(u) + S_{(i,j),\text{sup}}^{(-)}(u) \quad (40a)$$

$$S_{(i,j),\text{sup}}^{(+)}(u) = \delta_j \frac{g_i}{g_j} u \sigma_{i,j}(u) f(u - V_{i,j}) \quad (40b)$$

$$S_{(i,j),\text{sup}}^{(-)}(u) = -\delta_j \frac{g_i}{g_j} (u + V_{i,j}) \sigma_{i,j}(u + V_{i,j}) f(u) = -S_{(i,j),\text{sup}}^{(+)}(u + V_{i,j}) . \quad (40c)$$

As indicated above, the discretization monitors the shift of an electron due to excitations with threshold $V_{i,j}$, from lower energies starting at $u_n^{\text{node}} - V_{i,j} \geq u_{m_n}^{\text{node}}$ into a higher-energy interval $\Delta u_n = u_{n+1}^{\text{node}} - u_n^{\text{node}}$.

Because $\Delta u_n \geq \Delta u_{m_n}$ (again, the equality holds in the case of an uniform energy-grid), it is possible that the large interval Δu_n contains several small intervals Δu_{m_n+p} , with $p = 1, 2, \dots, \mathcal{M}$, the situation becoming “critical” for large thresholds that might deviate considerably u_n^{node} from $u_{m_n}^{\text{node}}$ (see figure 4).

In this case the operator is discretized as follows, again with focus on its entrance component into intervals

$$u_{m_n}^{\text{node}} \leq u_n^{\text{node}} - V_{i,j} < u_{m_n+1}^{\text{node}} < u_{m_n+2}^{\text{node}} < \dots < u_{m_n+\mathcal{M}+1}^{\text{node}} \leq u_{n+1}^{\text{node}} - V_{i,j} .$$

¹⁰Here, we take the first expression in brackets of equation (38b). Indeed, in an uniform energy-grid, the second expression is identical to the first one, since $u_{n+1}^{\text{node}} - (u_{m_n+1}^{\text{node}} - V_{i,j}) = u_{n+1}^{\text{node}} + V_{i,j} - u_{m_n+1}^{\text{node}} = u_{n+l_{i,j}+1} - u_{m_n+1} = 0$.

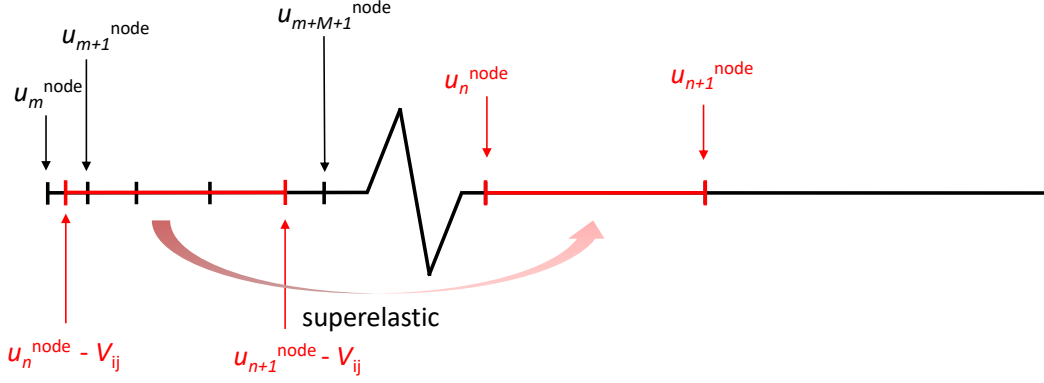


Figure 4: Graphic representation of a generic superelastic collision in a non-uniform energy grid.

$$\begin{aligned}
S_n^{\text{sup}} &= \sum_{i,j>i} S_{(i,j)_n}^{\text{sup}} = \sum_{i,j>i} [S_{(i,j)_n}^{\text{sup},(+)} + S_{(i,j)_n}^{\text{sup},(-)}] \\
S_{(i,j)_n}^{\text{sup},(+)} \Delta u_n &= \int_{u_n^{\text{node}}}^{u_{n+1}^{\text{node}}} S_{(i,j)}^{\text{sup},(+)}(u) du \\
&= \int_{u_n^{\text{node}} - V_{i,j}}^{u_{n+1}^{\text{node}} - V_{i,j}} S_{(i,j)}^{\text{sup},(+)}(u + V_{i,j}) du \\
&= \int_{u_n^{\text{node}} - V_{i,j}}^{u_{m_n+1}^{\text{node}}} S_{(i,j)}^{\text{sup},(+)}(u + V_{i,j}) du + \\
&\quad (1 - \delta_{\mathcal{M},0}) \sum_{p=1}^{\mathcal{M}} \int_{u_{m_n+1+(p-1)}^{\text{node}}}^{u_{m_n+1+p}^{\text{node}}} S_{(i,j)}^{\text{sup},(+)}(u + V_{i,j}) du + \\
&\quad \int_{u_{m_n+1+\mathcal{M}}^{\text{node}}}^{u_{n+1}^{\text{node}} - V_{i,j}} S_{(i,j)}^{\text{sup},(+)}(u + V_{i,j}) du \\
&= \delta_j \frac{g_i}{g_j} \left[\int_{u_n^{\text{node}} - V_{i,j}}^{u_{m_n+1}^{\text{node}}} (u + V_{i,j}) \sigma_{i,j}(u + V_{i,j}) f(u) du + \right. \\
&\quad (1 - \delta_{\mathcal{M},0}) \sum_{p=1}^{\mathcal{M}} \int_{u_{m_n+1+(p-1)}^{\text{node}}}^{u_{m_n+1+p}^{\text{node}}} (u + V_{i,j}) \sigma_{i,j}(u + V_{i,j}) f(u) du + \\
&\quad \left. \int_{u_{m_n+1+\mathcal{M}}^{\text{node}}}^{u_{n+1}^{\text{node}} - V_{i,j}} (u + V_{i,j}) \sigma_{i,j}(u + V_{i,j}) f(u) du \right] \\
&= \delta_j \frac{g_i}{g_j} \left\{ \sigma_{(i,j)_n} f_{m_n} \left[\frac{1}{2} [(u_n^{\text{node}} - V_{i,j}) + u_{m_n+1}^{\text{node}}] + V_{i,j} \right] [u_{m_n+1}^{\text{node}} - (u_n^{\text{node}} - V_{i,j})] + \right. \\
&\quad (1 - \delta_{\mathcal{M},0}) \sum_{p=1}^{\mathcal{M}} \sigma_{(i,j)_n} f_{m_n+p} \left[\frac{1}{2} (u_{m_n+p}^{\text{node}} + u_{m_n+p+1}^{\text{node}}) + V_{i,j} \right] (u_{m_n+p+1}^{\text{node}} - u_{m_n+p}^{\text{node}}) + \\
&\quad \left. \sigma_{(i,j)_n} f_{m_n+\mathcal{M}+1} \left[\frac{1}{2} [u_{m_n+\mathcal{M}+1}^{\text{node}} + (u_{n+1}^{\text{node}} - V_{i,j})] + V_{i,j} \right] [(u_{n+1}^{\text{node}} - V_{i,j}) - u_{m_n+\mathcal{M}+1}^{\text{node}}] \right\} \\
&= \delta_j \frac{g_i}{g_j} \left\{ \sigma_{(i,j)_n} f_{m_n} \frac{1}{2} [u_n^{\text{node}} + (u_{m_n+1}^{\text{node}} + V_{i,j})] [(u_{m_n+1}^{\text{node}} + V_{i,j}) - u_n^{\text{node}}] + \right. \\
&\quad \left. (1 - \delta_{\mathcal{M},0}) \sum_{p=1}^{\mathcal{M}} \sigma_{(i,j)_n} f_{m_n+p} \times \right.
\end{aligned} \tag{41a}$$

$$\begin{aligned}
& \frac{1}{2} [(u_{m_n+p}^{\text{node}} + V_{i,j}) + (u_{m_n+p+1}^{\text{node}} + V_{i,j})] [(u_{m_n+p+1}^{\text{node}} + V_{i,j}) - (u_{m_n+p}^{\text{node}} + V_{i,j})] + \\
& \sigma_{(i,j)_n} f_{m_n+\mathcal{M}+1} \frac{1}{2} [(u_{m_n+\mathcal{M}+1} + V_{i,j}) + u_{n+1}^{\text{node}}] [u_{n+1}^{\text{node}} - (u_{m_n+\mathcal{M}+1} + V_{i,j})] \Big\} \\
= & \delta_j \frac{g_i}{g_j} \left\{ \sigma_{(i,j)_n} f_{m_n} \frac{1}{2} \left[(u_{c_{m_n+1}}^{i,j})^2 - (u_n^{\text{node}})^2 \right] + \right. \\
& (1 - \delta_{\mathcal{M},0}) \sum_{p=1}^{\mathcal{M}} \sigma_{(i,j)_n} f_{m_n+p} \frac{1}{2} \left[(u_{c_{m_n+p+1}}^{i,j})^2 - (u_{c_{m_n+p}}^{i,j})^2 \right] + \\
& \left. \sigma_{(i,j)_n} f_{m_n+\mathcal{M}+1} \frac{1}{2} \left[(u_{n+1}^{\text{node}})^2 - (u_{c_{m_n+\mathcal{M}+1}}^{i,j})^2 \right] \right\} , \tag{41b}
\end{aligned}$$

with $\delta_{\mathcal{M},0}$ the Kronecker symbol and where we have used equation (38c).
Moreover,

$$\begin{aligned}
S_{(i,j)_n}^{\text{sup},(+)} \Delta u_n &= \int_{u_n^{\text{node}}}^{u_{n+1}^{\text{node}}} S_{(i,j)}^{\text{sup},(+)}(u) du \\
&= - \int_{u_n^{\text{node}}}^{u_{n+1}^{\text{node}}} S_{(i,j)}^{\text{sup},(-)}(u - V_{i,j}) du \\
&= - \int_{u_n^{\text{node}} - V_{i,j}}^{u_{n+1}^{\text{node}} - V_{i,j}} S_{(i,j)}^{\text{sup},(-)}(u) du \\
&\equiv - \sum_{k=m_n}^{m_n+\mathcal{M}+1} S_{(i,j)_k}^{\text{sup},(-)} \Delta u_k , \tag{41c}
\end{aligned}$$

since electrons coming out from all cells $m_n, m_n + 1, \dots, m_n + \mathcal{M} + 1$ will enter the single cell n .

Finally

$$\begin{aligned}
S_{(i,j)_n}^{\text{sup},(+)} &\equiv \sum_{k=m_n}^{m_n+\mathcal{M}+1} C_{n,k}^{(i,j),\text{sup}} f_k = \delta_j \frac{g_i}{g_j} \left\{ \frac{(u_{c_{m_n+1}}^{i,j})^2 - (u_n^{\text{node}})^2}{2\Delta u_n} \sigma_{(i,j)_n} f_{m_n} + \right. \\
& (1 - \delta_{\mathcal{M},0}) \sum_{p=1}^{\mathcal{M}} \frac{(u_{c_{m_n+p+1}}^{i,j})^2 - (u_{c_{m_n+p}}^{i,j})^2}{2\Delta u_n} \sigma_{(i,j)_n} f_{m_n+p} + \\
& \left. \frac{(u_{n+1}^{\text{node}})^2 - (u_{c_{m_n+\mathcal{M}+1}}^{i,j})^2}{2\Delta u_n} \sigma_{(i,j)_n} f_{m_n+\mathcal{M}+1} \right\} \tag{42a}
\end{aligned}$$

$$S_{(i,j)_k}^{\text{sup},(-)} \equiv C_{k,k}^{(i,j),\text{sup}} f_k = -C_{n,k}^{(i,j),\text{sup}} \frac{\Delta u_n}{\Delta u_k} f_k , \text{ for } k = m_n, m_n + 1, \dots, m_n + \mathcal{M} + 1 \tag{42b}$$

$$\begin{aligned}
\Rightarrow \sum_n S_{(i,j)_n}^{\text{sup},(+)} \Delta u_n &= \sum_n \sum_{k=m_n}^{m_n+\mathcal{M}+1} C_{n,k}^{(i,j),\text{sup}} f_k \Delta u_n \\
&= - \sum_n \sum_{k=m_n}^{m_n+\mathcal{M}+1} C_{k,k}^{(i,j),\text{sup}} f_k \Delta u_k = - \sum_n \sum_{k=m_n}^{m_n+\mathcal{M}+1} S_{(i,j)_k}^{\text{sup},(-)} \Delta u_k , \tag{42c}
\end{aligned}$$

for $n, m_n, k = 1, 2, \dots, \mathcal{N}$.

Equations (42a) use the Klein-Rosseland relation to express the superelastic cross sections as function of the corresponding inelastic. An alternative expression, using directly the superelastic cross sections,

is (is LMarques' code using this alternative expression ?)

$$S_{(i,j)_n}^{\text{sup},(+)} \equiv \sum_{k=m_n}^{m_n+\mathcal{M}+1} C_{n,k}^{(i,j),\text{sup}} f_k = \delta_j \left\{ \frac{(u_{m_n+1}^{\text{node}})^2 - (u_{d_n}^{i,j})^2}{2\Delta u_n} \sigma_{(i,j)_{m_n}} f_{m_n} + \right. \\ (1 - \delta_{\mathcal{M},0}) \sum_{p=1}^{\mathcal{M}} \frac{(u_{m_n+p+1}^{\text{node}})^2 - (u_{m_n+p}^{\text{node}})^2}{2\Delta u_n} \sigma_{(i,j)_{m_n+p}} f_{m_n+p} + \\ \left. \frac{(u_{d_{n+1}}^{i,j})^2 - (u_{m_n+\mathcal{M}+1}^{\text{node}})^2}{2\Delta u_n} \sigma_{(i,j)_{m_n+\mathcal{M}+1}} f_{m_n+\mathcal{M}+1} \right\} , \quad (43a)$$

with

$$u_{d_n}^{i,j} \equiv u_n^{\text{node}} - V_{ij} . \quad (43b)$$

Similarly to the inelastic operator, and after considering the same methodologic differences in the discretization, the previous expressions allow retrieving the usual equation of the discretized superelastic operator in a uniform energy grid.

$$S_{(i,j)_n}^{\text{sup},(+)} \Delta u_n = -S_{(i,j)_{m_n}}^{\text{sup},(-)} \Delta u_{m_n} = \delta_j \frac{g_i}{g_j} \sigma_{(i,j)_n} f_{m_n} \left[\frac{1}{2} (u_n^{\text{node}} + u_{n+1}^{\text{node}}) \right] (u_{n+1}^{\text{node}} - u_n^{\text{node}}) \\ \Rightarrow S_{(i,j)_n}^{\text{sup},(+)} = -S_{(i,j)_{n-l_{ij}}}^{\text{sup},(-)} = \delta_i \frac{g_i}{g_j} u_n \sigma_{(i,j)_n} f_{n-l_{ij}} \\ \Rightarrow S_{(i,j)_n}^{\text{sup}} = S_{(i,j)_n}^{\text{sup},(+)} + S_{(i,j)_n}^{\text{sup},(-)} = \delta_i \frac{g_i}{g_j} \left[u_n \sigma_{(i,j)_n} f_{n-l_{ij}} - u_{n+l_{ij}} \sigma_{(i,j)_{n+l_{ij}}} f_n \right]$$

– Ionization collision operator

Check against LMarques' code

The ionization collision operator is also separated into its gain/loss components (see equation (7a)),

$$S_{i,\text{ion}}(u) = S_{i,\text{ion}}^{(+),\text{sec}}(u) + S_{i,\text{ion}}^{(+),\text{scat}}(u) + S_{i,\text{ion}}^{(-)}(u) \quad (44a)$$

$$S_{i,\text{ion}}^{(+),\text{sec}}(u) = \delta_i \int_{2u+V_{i,\text{ion}}}^{\infty} u' \sigma_{i,\text{ion}}^{\text{sec}}(u', u) f(u') du' \quad (44b)$$

$$S_{i,\text{ion}}^{(+),\text{scat}}(u) = \delta_i \int_{u+V_{i,\text{ion}}}^{2u+V_{i,\text{ion}}} u' \sigma_{i,\text{ion}}^{\text{sec}}(u', u' - V_{i,\text{ion}} - u) f(u') du' \quad (44c)$$

$$S_{i,\text{ion}}^{(-)}(u) = -\delta_i u \sigma_{i,\text{ion}}(u) f(u) = -\delta_i u f(u) \int_0^{(u-V_{i,\text{ion}})/2} \sigma_{i,\text{ion}}^{\text{sec}}(u, u') du' . \quad (44d)$$

In the previous equations:

- * $S_{i,\text{ion}}^{(+),\text{sec}}(u)$ represents the gain of *secondary* electrons created at energy u from *incident* (or primary) electrons with energy u' ;
- * $S_{i,\text{ion}}^{(+),\text{scat}}(u)$ represents the gain of *scattered* electrons created at the remainder energy $u_s = u' - V_{i,\text{ion}} - u$ from *incident* electrons with energy u' , such that $u' = V_{i,\text{ion}} + u + u_s$;
- * $S_{i,\text{ion}}^{(-)}(u)$ represents the loss of *incident* electrons with energy u , yielding *secondary* electrons with energy $0 \leq u' \leq (u - V_{i,\text{ion}})/2$ and (consequently) *scattered* electrons with energy $(u - V_{i,\text{ion}})/2 \leq u_s \leq u - V_{i,\text{ion}}$.

In the case of the ionization operator, neither the secondary electron nor the scattered electron have uniquely defined energies, and therefore the discretization starts by defining the loss of primary electrons with (higher) energies u' , defined by the corresponding loss component $S_{i,\text{ion}}^{(-)}(u')$.

$$\begin{aligned}
S_{(i)_n}^{\text{ion},(-)} \Delta u_n &= \int_{u_n^{\text{node}}}^{u_{n+1}^{\text{node}}} S_{(i)}^{\text{ion},(-)}(u') du' \\
&= -\delta_i \int_{u_n^{\text{node}}}^{u_{n+1}^{\text{node}}} \left[u' f(u') \int_0^{(u' - V_{i,\text{ion}})/2} \sigma_{i,\text{ion}}^{\text{sec}}(u', u) du \right] du' \quad (45a)
\end{aligned}$$

In this case the discretization monitors the shift of an electron due to deexcitations with threshold $V_{i,\text{ion}}$, from higher energies starting at $u_n^{\text{node}} + V_{i,j} \leq u_{m_{n+1}}^{\text{node}}$ into a lower-energy interval $\Delta u_n = u_{n+1}^{\text{node}} - u_n^{\text{node}}$ (see figure 3).

- **Attachment collision operator**
TO CHECK / UPDATE

$$S_n = \sum_{i,j>i} S_{(i,j)_n} + \sum_i S_{(i,\text{ion})_n} + \sum_i S_{(i,\text{att})_n} \quad (46a)$$

$$\sigma_{i,\text{ion}}(u_n^{\text{cell}}) = \sum_{m=1}^{(n-l_{i,\text{ion}})/2} \sigma_{(i,\text{ion})_{(n,m)}}^{\text{sec}} \quad (46b)$$

$$\begin{aligned}
S_{(i,\text{ion})_n} &= \delta_i \left\{ \sum_{m=n+l_{i,\text{ion}}+1}^{2n+l_{i,\text{ion}}} u_m \sigma_{(i,\text{ion})_{(m,m-n-l_{i,\text{ion}})}}^{\text{sec}} f_m \Delta u \right. \\
&\quad \left. + \sum_{m=2n+l_{i,\text{ion}}+1}^{\mathcal{N}} u_m \sigma_{(i,\text{ion})_{(m,n)}}^{\text{sec}} f_m \Delta u - u_n \frac{\sigma_{(i,\text{ion})_n} + \sigma_{(i,\text{ion})_{n+1}}}{2} f_n \right\} \quad (47a)
\end{aligned}$$

$$S_{(i,\text{att})_n} = -\delta_i u_n \frac{\sigma_{(i,\text{att})_n} + \sigma_{(i,\text{att})_{n+1}}}{2} f_n \quad (47b)$$

- **Relevant expressions to use in evaluating the discrete operators**

The discrete operators presented above require defining the following quantities:

- cross sections at grid-cells n , which are defined using the interpolation formula (18)

$$\sigma_{(i,j)_n} = \frac{1}{2} \left(\sigma_{(i,j)_{n+1}}^{\text{node}} + \sigma_{(i,j)_n}^{\text{node}} \right) \quad ; \quad (48)$$

- the cell-number (*e.g.* m_n) where the electron energy has a certain value (*e.g.* $u_{c_n}^{i,j}$), which can be calculated using equation (20)

LMarques uses an interpolation/dichotomy method instead of this formula - to check

$$m_n = \text{INT} \left\{ \frac{\ln \left[\frac{u_{c_n}^{i,j}}{h} (a-1) + 1 \right]}{\ln a} + 1 \right\} \quad . \quad (49)$$

The previous equations are subjected to the following conditions:

- the flux boundary conditions $G(0) = G(u_{\max}) = 0$, which imply $g_{x_1}^{\text{node}} = g_{x_{\mathcal{N}+1}}^{\text{node}} = 0$, $u_1^{\text{node}} f_1^{1,\text{node}} = u_{\mathcal{N}+1}^{\text{node}} f_{\mathcal{N}+1}^{1,\text{node}} = 0$ (since $G_E(u) \propto u f^1(u)$, see equation (6a)), and $A_0 = A_{\mathcal{N}} = B_1 = B_{\mathcal{N}+1} = 0$ (see section 2.5);
- the imposition of a Maxwellian EEDF for $dG_{ee}/du = 0$, which is ensured by setting $a_{n,m} = \sqrt{(w_n/w_{m-1})(\Delta u_{m-1}/\Delta u_n) \exp((1/T_e)[(\Delta u_{m-1}/2w_{m-1}) - (\Delta u_n/2w_n)])} a_{n,m} \times a_{m-1,n+1}$ (see section 2.3.2).
- the energy conservation condition for dG_{ee}/du , which yields $b_{m,n} = (w_{m-1}/w_n)(\Delta u_n/\Delta u_{m-1}) a_{n,m}$ (see section 2.3.1).

2.3 Conservation conditions for the electron-electron collision operator

2.3.1 Energy conservation for electron-electron collisions

Electron-electron collisions should not affect the net gain/loss of energy by the electrons, which means that

$$\begin{aligned}\frac{\Theta_{ee}}{N} &= -\frac{1}{N} \int_0^\infty u \frac{dG_{ee}(u)}{du} du = \frac{1}{N} \int_0^\infty G_{ee}(u) du \\ &= -\gamma \int_0^\infty \left[g_{ee}(u) \left(I(u)f(u) + J(u) \frac{df(u)}{du} \right) \right] du \\ &= -\gamma g_{ee} \left\{ \int_0^\infty \left[I(u)f(u) - \frac{dJ(u)}{du} f(u) \right] du + [J(u)f(u)]_0^\infty \right\} = 0 \quad ,\end{aligned}\tag{50a}$$

where in the last expression we have used the fact that

$$g_{ee} \propto \frac{u^{3/2}}{v^3} = \text{const} \quad ,\tag{50b}$$

and performed an integration by parts of the last term.

The Spitzer integral J , after an integration by parts, can be written as (see (34c))

$$J(u) = \frac{2}{3} \left[\int_0^u f(u') u'^{3/2} du' + u^{3/2} \int_u^\infty f(u') du' \right] = \int_0^u u'^{1/2} \left[\int_{u'}^\infty f(u'') du'' \right] du' \quad ,\tag{50c}$$

which yields

$$\frac{dJ(u)}{du} = u^{1/2} \int_u^\infty f(u') du' \tag{50d}$$

$$[J(u)f(u)]_0^\infty = \int_0^\infty u^{1/2} \left[\int_{u'}^\infty f(u'') du'' \right] du' \times f(\infty) - 0 \longrightarrow 0 \quad ,\tag{50e}$$

where we have used Leibnitz law and the fact that the EEDF vanishes at ∞ .

Moreover, noting that the derivative of the Spitzer integral I yields (see (34b))

$$I(u) = \int_0^u f(u') u'^{1/2} du' \implies \frac{dI(u)}{du} = f(u) u^{1/2} \quad ,\tag{50f}$$

one can write using (50d),

$$\begin{aligned}\int_0^\infty \frac{dJ(u)}{du} f(u) du &= \int_0^\infty f(u) u^{1/2} \int_u^\infty f(u') du' du \\ &= \int_0^\infty \frac{dI(u)}{du} \int_u^\infty f(u') du' du \\ &= \left[I(u) \int_u^\infty f(u') du' \right]_0^\infty + \int_0^\infty I(u) f(u) du \quad ,\end{aligned}\tag{50g}$$

where the last expression was obtained with an integration by parts.

Finally, using (50e) and (50g) in (50a) one obtains

$$\begin{aligned}\frac{\Theta_{ee}}{N} &= -\gamma g_{ee} \left\{ \int_0^\infty [I(u)f(u) - I(u)f(u)] du - \left[I(u) \int_u^\infty f(u') du' \right]_0^\infty \right\} \\ &= \gamma g_{ee} \left\{ \int_0^\infty f(u) u^{1/2} du \times 0 - 0 \times \int_0^\infty f(u) du \right\} \longrightarrow 0 \quad ,\end{aligned}\tag{50h}$$

thus confirming the result of equation (50a).

Numerically, this result introduces a relationship between the matrix terms of the electron-electron collision operator. From (69e), one has

$$\begin{aligned} \gamma \sum_{n=1}^{\mathcal{N}} \left[\frac{\Delta u_n}{2w_n} A_n - \frac{\Delta u_{n-1}}{2w_{n-1}} B_n \right] f_n = 0 &\implies \gamma \sum_{n=1}^{\mathcal{N}} \left[\frac{\Delta u_n}{2w_n} \sum_{m=1}^{\mathcal{N}} a_{n,m} - \frac{\Delta u_{n-1}}{2w_{n-1}} \sum_{m=1}^{\mathcal{N}} b_{n,m} \right] f_m f_n = 0 \\ &\implies \sum_{n,m=1}^{\mathcal{N}} \left[\frac{\Delta u_n}{2w_n} a_{n,m} - \frac{\Delta u_{n-1}}{2w_{n-1}} b_{n,m} \right] f_m f_n = 0 \quad , \end{aligned} \quad (51a)$$

which means that the matrix multiplying the EEDFs should be antisymmetric satisfying

$$\frac{\Delta u_n}{2w_n} a_{n,m} - \frac{\Delta u_{n-1}}{2w_{n-1}} b_{n,m} = -\frac{\Delta u_m}{2w_m} a_{m,n} + \frac{\Delta u_{m-1}}{2w_{m-1}} b_{m,n} \quad (51b)$$

or

$$\begin{aligned} \frac{\Delta u_n}{2w_n} a_{n,m} - \frac{\Delta u_{m-1}}{2w_{m-1}} b_{m,n} &= -\frac{\Delta u_m}{2w_m} a_{m,n} + \frac{\Delta u_{n-1}}{2w_{n-1}} b_{n,m} = 0 \\ &\implies b_{m,n} = \frac{w_{m-1}}{w_n} \frac{\Delta u_n}{\Delta u_{m-1}} a_{n,m} \quad . \end{aligned} \quad (51c)$$

2.3.2 Maxwellian solution for collisional equilibrium

Under the exclusive action of electron-electron collisions, the EBE (2a), (3a) or (4a) writes

$$\begin{aligned} \frac{dG_{ee}(u)}{du} = 0 &\implies G_{ee}(u) = 0 \\ &\implies g_{ee}(u) \left(I(u)f(u) + J(u) \frac{df(u)}{du} \right) = 0 \quad , \end{aligned} \quad (52a)$$

where we have used the boundary conditions $G_{ee}(0) = G_{ee}(\infty) = 0$ and equation (35).

Equation (52a) can be rewritten using (34b), (50b) and (50c) to obtain

$$f(u) \int_0^u u'^{1/2} f(u') du' + \frac{df(u)}{du} \int_0^u u'^{1/2} \left[\int_{u'}^{\infty} f(u'') du'' \right] du' = 0 \quad . \quad (52b)$$

This identity is satisfied if f is a Maxwellian distribution function, since in this case

$$f(u) = a \int_u^{\infty} f(u') du' \implies \frac{df(u)}{du} = -af(u) \implies f(u) = C \exp(-au) \quad , \quad (52c)$$

with $a \equiv 1/T_e$ and C an arbitrary constant.

Numerically, this equilibrium situation implies that (see (35) and (51c))

$$\begin{aligned}
& \frac{1}{\Delta u_n} \left[\sum_{m=1}^{\mathcal{N}} (a_{n-1,m} f_m) f_{n-1} - \sum_{m=1}^{\mathcal{N}} (a_{n,m} f_m + b_{n,m} f_m) f_n + \sum_{m=1}^{\mathcal{N}} (b_{n+1,m} f_m) f_{n+1} \right] = 0 \\
& \Rightarrow \sum_{m=1}^{\mathcal{N}} (a_{n-1,m} f_m) f_{n-1} - \sum_{m=1}^{\mathcal{N}} \left(a_{n,m} f_m + \frac{w_{n-1}}{w_m} \frac{\Delta u_m}{\Delta u_{n-1}} a_{m,n} f_m \right) f_n \\
& \quad + \sum_{m=1}^{\mathcal{N}} \left(\frac{w_n}{w_m} \frac{\Delta u_m}{\Delta u_n} a_{m,n+1} f_m \right) f_{n+1} = 0 \\
& \Rightarrow \sum_{m=1}^{\mathcal{N}} a_{n-1,m} f_m f_{n-1} - \sum_{m=1}^{\mathcal{N}} a_{n,m} f_m f_n - \sum_{m=1}^{\mathcal{N}-1} \frac{w_{n-1}}{w_m} \frac{\Delta u_m}{\Delta u_{n-1}} a_{m,n} f_m f_n \\
& \quad + \sum_{m=1}^{\mathcal{N}-1} \frac{w_n}{w_m} \frac{\Delta u_m}{\Delta u_n} a_{m,n+1} f_m f_{n+1} = 0 \tag{53a} \\
& \Rightarrow \sum_{m=1}^{\mathcal{N}} a_{n-1,m} f_m f_{n-1} - \sum_{m=2}^{\mathcal{N}} \frac{w_{n-1}}{w_{m-1}} \frac{\Delta u_{m-1}}{\Delta u_{n-1}} a_{m-1,n} f_{m-1} f_n \\
& \quad - \sum_{m=1}^{\mathcal{N}} a_{n,m} f_m f_n + \sum_{m=2}^{\mathcal{N}} \frac{w_n}{w_{m-1}} \frac{\Delta u_{m-1}}{\Delta u_n} a_{m-1,n+1} f_{m-1} f_{n+1} = 0 \\
& \Rightarrow \sum_{m=1}^{\mathcal{N}} \left[a_{n-1,m} f_m f_{n-1} - \frac{w_{n-1}}{w_{m-1}} \frac{\Delta u_{m-1}}{\Delta u_{n-1}} a_{m-1,n} f_{m-1} f_n \right] \\
& \quad - \sum_{m=1}^{\mathcal{N}} \left[a_{n,m} f_m f_n - \frac{w_n}{w_{m-1}} \frac{\Delta u_{m-1}}{\Delta u_n} a_{m-1,n+1} f_{m-1} f_{n+1} \right] = 0 \quad ,
\end{aligned}$$

where we have used the boundary conditions $A_0 = A_{\mathcal{N}} = 0$ or, equivalently, $a_{0,k} = a_{\mathcal{N},k} = 0$, and where the EEDFs are Maxwellian distribution functions satisfying (see equation (19))

$$\begin{aligned}
f_{m-1} f_{n+1} &= C^2 \exp(-a u_{m-1}) \exp(-a u_{n+1}) \\
&= C^2 \exp(-a u_m) \exp(-a u_n) \exp\left(a \frac{\Delta u_{m-1}}{2 w_{m-1}}\right) \exp\left(-a \frac{\Delta u_n}{2 w_n}\right) \\
&= f_m f_n \exp\left(a \frac{\Delta u_{m-1}}{2 w_{m-1}}\right) \exp\left(-a \frac{\Delta u_n}{2 w_n}\right) \quad . \tag{53b}
\end{aligned}$$

Equations (53a)-(53b) allow defining a relationship between the upper/lower matrix terms a_{mn} of the electron-electron collision operator

$$a_{n,m} = \frac{w_n}{w_{m-1}} \frac{\Delta u_{m-1}}{\Delta u_n} \exp\left(\frac{1}{T_e} \left[\frac{\Delta u_{m-1}}{2 w_{m-1}} - \frac{\Delta u_n}{2 w_n} \right]\right) a_{m-1,n+1}, \quad 1 \leq n \leq \mathcal{N}-1 \quad ; \quad 1 \leq m \leq \mathcal{N}$$

or to balance out the contributions from the upper/lower matrix terms

$$a_{n,m} = \sqrt{\frac{w_n}{w_{m-1}} \frac{\Delta u_{m-1}}{\Delta u_n} \exp\left(\frac{1}{T_e} \left[\frac{\Delta u_{m-1}}{2 w_{m-1}} - \frac{\Delta u_n}{2 w_n} \right]\right) a_{n,m} \times a_{m-1,n+1}}, \quad 1 \leq n \leq \mathcal{N}-1 \quad ; \quad 1 \leq m \leq \mathcal{N}, \tag{54}$$

with T_e calculated from equation (36g).

2.4 The particle balance equation

Equations (2a)-(4b) can be integrated in energy space to yield the particle balance equation, *i.e.*

- For a PT situation

$$\frac{1}{\gamma} \frac{\langle \nu_{\text{eff}} \rangle}{N} \int_0^\infty \sqrt{u} f(u) du + \frac{1}{N} \frac{1}{\gamma} \int_0^\infty \frac{dG(u)}{du} du = \int_0^\infty S(u) du$$

and therefore

$$\frac{\langle \nu_{\text{eff}} \rangle}{N} = \sqrt{\frac{2e}{m_e}} \int_0^\infty S(u) du \quad , \quad (55)$$

where we have used the normalization condition and the boundary conditions for the upflux $G(u)$.

- For a SST situation

$$-\frac{1}{3} \left(\frac{\alpha_{\text{eff}}}{N} \right)^2 \int_0^\infty \frac{u}{\Omega_{\text{SST}}(u)} f(u) du - \frac{1}{3} \frac{\alpha_{\text{eff}}}{N} \frac{E}{N} \int_0^\infty \frac{u}{\Omega_{\text{SST}}(u)} \frac{df(u)}{du} du + \frac{1}{N} \frac{1}{\gamma} \int_0^\infty \frac{dG(u)}{du} du = \int_0^\infty S(u) du$$

and therefore

$$(D_e N) \left(\frac{\alpha_{\text{eff}}}{N} \right)^2 - (\mu_e N) \frac{E}{N} \frac{\alpha_{\text{eff}}}{N} = \frac{\langle \nu_{\text{eff}} \rangle}{N} \quad , \quad (56)$$

where we have used equation (58), the boundary conditions for the upflux $G(u)$, and the expressions of the DC reduced transverse diffusion coefficient and the DC reduced mobility (in $\text{m}^{-1} \text{s}^{-1}$ and $\text{eV}^{-1} \text{m}^{-1} \text{s}^{-1}$, respectively),

$$D_e N = \frac{\gamma}{3} \int_0^\infty \frac{u}{\Omega_{\text{SST}}(u)} f(u) du \quad (57a)$$

$$\mu_e N = -\frac{\gamma}{3} \int_0^\infty \frac{u}{\Omega_{\text{SST}}(u)} \frac{df(u)}{du} du \quad . \quad (57b)$$

- For a time-dependent situation

$$\frac{1}{N} \frac{1}{\gamma} \frac{d}{dt} \int_0^\infty \sqrt{u} f(u, t) du + \frac{1}{\gamma} \frac{\langle \nu_{\text{eff}} \rangle(t)}{N} \int_0^\infty \sqrt{u} f(u, t) du + \frac{1}{N} \frac{1}{\gamma} \int_0^\infty \frac{\partial G(u, t)}{\partial u} du = \int_0^\infty S(u, t) du$$

and therefore

$$\frac{\langle \nu_{\text{eff}} \rangle(t)}{N} = \sqrt{\frac{2e}{m_e}} \int_0^\infty S(u, t) du \quad , \quad (58)$$

where, similarly to the PT situation, we have used the normalization condition and the boundary conditions for the upflux $G(u)$.

Note that the integration of the right-hand sides of equations (2a), (3a) and (4a) yields

$$\begin{aligned} \int_0^\infty S_{\text{inel}}(u) du &= \sum_{i,j>i} \int_0^\infty S_{(i,j),\text{inel}}(u) du = \\ &= \sum_{i,j>i} \int_0^\infty \delta_i [\sigma_{i,j}(u + V_{i,j}) f(u + V_{i,j})(u + V_{i,j}) - \sigma_{i,j}(u) f(u) u] du \\ &= \sum_{i,j>i} \delta_i \left[\int_{V_{i,j}}^\infty \sigma_{i,j}(u') f(u') u' du' - \int_0^\infty \sigma_{i,j}(u) f(u) u du \right] \\ &= \sum_{i,j>i} \delta_i \left[\int_0^\infty \sigma_{i,j}(u) f(u) u du - \int_0^\infty \sigma_{i,j}(u) f(u) u du \right] \\ &= 0 \quad , \end{aligned} \quad (59)$$

where we have considered that $\sigma_{i,j}(u) = 0$ for $u \leq V_{i,j}$.

The numerical discretization of the EBE must ensure the electron-particle conservation, *i.e.* it must allow retrieving the numerical version of equations (58)-(56). In particular:

- normalization condition

The EEDF is calculated as to satisfy the normalization condition

$$\int_0^\infty f(u) \sqrt{u} du = 1 \longrightarrow \sum_{n=1}^{\mathcal{N}} f_n \sqrt{u_n} \Delta u_n = 1 \quad ; \quad (60)$$

- free-diffusion coefficient ¹¹

The DC reduced transverse diffusion coefficient is calculated using (see equations (26) and (17b); recall that $u_1^{\text{node}} f_1^{1,\text{node}} = u_{\mathcal{N}+1}^{\text{node}} f_{\mathcal{N}+1}^{1,\text{node}} = 0$)

$$\begin{aligned} D_e N &= \frac{\gamma}{3} \int_0^\infty \frac{u}{\Omega_{\text{SST}}(u)} f(u) du \\ &\longrightarrow \frac{\gamma}{3} \sum_{n=1}^{\mathcal{N}} \int_{u_n^{\text{node}}}^{u_{n+1}^{\text{node}}} \frac{u}{\Omega_{\text{SST}}(u)} f(u) du \\ &\longrightarrow \frac{\gamma}{3} \sum_{n=1}^{\mathcal{N}} \frac{1}{2} \left[\frac{u_{n+1}^{\text{node}}}{\Omega_{\text{SST}_{n+1}}^{\text{node}}} f_{n+1}^{\text{node}} + \frac{u_n^{\text{node}}}{\Omega_{\text{SST}_n}^{\text{node}}} f_n^{\text{node}} \right] \Delta u_n \\ &\longrightarrow \frac{\gamma}{3} \left[\sum_{n=2}^{\mathcal{N}+1} \frac{u_n^{\text{node}}}{\Omega_{\text{SST}_n}^{\text{node}}} f_n^{\text{node}} \frac{\Delta u_{n-1}}{2} + \sum_{n=1}^{\mathcal{N}} \frac{u_n^{\text{node}}}{\Omega_{\text{SST}_n}^{\text{node}}} f_n^{\text{node}} \frac{\Delta u_n}{2} \right] \\ &\longrightarrow \frac{\gamma}{3} \sum_{n=1}^{\mathcal{N}} \frac{u_n^{\text{node}}}{\Omega_{\text{SST}_n}^{\text{node}}} f_n^{\text{node}} \left(\frac{\Delta u_{n-1} + \Delta u_n}{2} \right) \\ &\longrightarrow \frac{\gamma}{3} \sum_{n=1}^{\mathcal{N}} \frac{u_n^{\text{node}}}{\Omega_{\text{SST}_n}^{\text{node}}} [(1 - w_{n-1}) f_{n-1} + w_{n-1} f_n] \left(\frac{\Delta u_{n-1} + \Delta u_n}{2} \right) \quad ; \end{aligned} \quad (61)$$

- mobility ¹²

The DC mobility is calculated using (see equation (26); recall that $u_1^{\text{node}} f_1^{1,\text{node}} = u_{\mathcal{N}+1}^{\text{node}} f_{\mathcal{N}+1}^{1,\text{node}} = 0$)

$$\begin{aligned} \mu_e N &= -\frac{\gamma}{3} \int_0^\infty \frac{u}{\Omega_{\text{SST}}(u)} \frac{df(u)}{du} du \\ &\longrightarrow -\frac{\gamma}{3} \sum_{n=1}^{\mathcal{N}} \int_{u_n^{\text{node}}}^{u_{n+1}^{\text{node}}} \frac{u}{\Omega_{\text{SST}}(u)} \frac{df(u)}{du} du \\ &\longrightarrow -\frac{\gamma}{3} \sum_{n=1}^{\mathcal{N}} \frac{1}{2} \left[\frac{u_{n+1}^{\text{node}}}{\Omega_{\text{SST}_{n+1}}^{\text{node}}} \left. \frac{df}{du} \right|_{n+1}^{\text{node}} + \frac{u_n^{\text{node}}}{\Omega_{\text{SST}_n}^{\text{node}}} \left. \frac{df}{du} \right|_n^{\text{node}} \right] \Delta u_n \\ &\longrightarrow -\frac{\gamma}{3} \sum_{n=1}^{\mathcal{N}} \frac{1}{2} \left[\frac{u_{n+1}^{\text{node}}}{\Omega_{\text{SST}_{n+1}}^{\text{node}}} \frac{f_{n+1} - f_n}{\Delta u_{n+1}/2 + \Delta u_n/2} + \frac{u_n^{\text{node}}}{\Omega_{\text{SST}_n}^{\text{node}}} \frac{f_n - f_{n-1}}{\Delta u_n/2 + \Delta u_{n-1}/2} \right] \Delta u_n \\ &\longrightarrow -\frac{\gamma}{3} \left[\sum_{n=2}^{\mathcal{N}+1} \frac{u_n^{\text{node}}}{\Omega_{\text{SST}_n}^{\text{node}}} \frac{f_n - f_{n-1}}{\Delta u_n + \Delta u_{n-1}} \Delta u_{n-1} + \sum_{n=1}^{\mathcal{N}} \frac{u_n^{\text{node}}}{\Omega_{\text{SST}_n}^{\text{node}}} \frac{f_n - f_{n-1}}{\Delta u_n + \Delta u_{n-1}} \Delta u_n \right] \\ &\longrightarrow -\frac{\gamma}{3} \sum_{n=1}^{\mathcal{N}} \frac{u_n^{\text{node}}}{\Omega_{\text{SST}_n}^{\text{node}}} (f_n - f_{n-1}) \quad ; \end{aligned} \quad (62)$$

- flux conservation

The integration of each upflux term yields a zero-identity (recall that $f_0 = f_{\mathcal{N}+1} = 0$ and $g_{x_1}^{\text{node}} = g_{x_{\mathcal{N}+1}}^{\text{node}} =$

¹¹The discretization of the free-diffusion coefficient adopts a framework different from that of LoKI-B v2.2.0, since it relates with the anisotropic function f^1 which is defined on the grid-nodes (not the grid-cells). Indeed, today we believe that the description of the free-diffusion coefficient in LoKI-B v2.2.0 is **incorrect**, since it is not consistent with that adopted in the power absorbed from the electric field.

¹²The discretization of the mobility adopts a framework different from that of LoKI-B v2.2.0, since it relates with the anisotropic function f^1 which is defined on the grid-nodes (not the grid-cells). Indeed, today we believe that the description of the mobility in LoKI-B v2.2.0 is **incorrect**, since it is not consistent with that adopted in the power absorbed from the electric field.

0; see equation (22b))

$$\begin{aligned}
-\frac{1}{N} \frac{1}{\gamma} \int_0^\infty \frac{dG_x(u)}{du} du &= 0 \longrightarrow -\frac{1}{N} \frac{1}{\gamma} \sum_{n=1}^{\mathcal{N}} \left(\frac{dG_x(u)}{du} \right)_n \Delta u_n \\
&= \sum_{n=1}^{\mathcal{N}} \left\{ \frac{g_{x_n}^{\text{node}}}{\Delta u_n} \left[c_x \frac{2w_{n-1}}{\Delta u_{n-1}} - d_x (1 - w_{n-1}) \right] \right\} f_{n-1} \Delta u_n \\
&\quad - \sum_{n=1}^{\mathcal{N}} \left\{ \frac{g_{x_{n+1}}^{\text{node}}}{\Delta u_n} \left[c_x \frac{2w_n}{\Delta u_n} - d_x (1 - w_n) \right] + \frac{g_{x_n}^{\text{node}}}{\Delta u_n} \left[c_x \frac{2w_{n-1}}{\Delta u_{n-1}} + d_x w_{n-1} \right] \right\} f_n \Delta u_n \\
&\quad + \sum_{n=1}^{\mathcal{N}} \left\{ \frac{g_{x_{n+1}}^{\text{node}}}{\Delta u_n} \left[c_x \frac{2w_n}{\Delta u_n} + d_x w_n \right] \right\} f_{n+1} \Delta u_n \\
&= \sum_{n=1}^{\mathcal{N}-1} \left\{ g_{x_{n+1}}^{\text{node}} \left[c_x \frac{2w_n}{\Delta u_n} - d_x (1 - w_n) \right] \right\} f_n \\
&\quad - \sum_{n=1}^{\mathcal{N}-1} \left\{ g_{x_{n+1}}^{\text{node}} \left[c_x \frac{2w_n}{\Delta u_n} - d_x (1 - w_n) \right] \right\} f_n - \sum_{n=2}^{\mathcal{N}} \left\{ g_{x_n}^{\text{node}} \left[c_x \frac{2w_{n-1}}{\Delta u_{n-1}} + d_x w_{n-1} \right] \right\} f_n \\
&\quad + \sum_{n=2}^{\mathcal{N}} \left\{ g_{x_n}^{\text{node}} \left[c_x \frac{2w_{n-1}}{\Delta u_{n-1}} + d_x w_{n-1} \right] \right\} f_n \\
&= 0 \quad ;
\end{aligned} \tag{63}$$

- particle conservation in inelastic/superelastic collisions

The integration of the inelastic and superelastic collision terms yields a zero identify (see equations (39a)-(39c) and (42a)-(42c))

$$\begin{aligned}
\int_0^\infty S_{\text{inel}}(u) du &= 0 \longrightarrow \sum_{n=1}^{\mathcal{N}} S_n^{\text{inel}} \Delta u_n = \sum_{i,j>i} \sum_{n=1}^{\mathcal{N}} S_{(i,j)_n}^{\text{inel}} \Delta u_n \\
&= \sum_{i,j>i} \sum_{n=1}^{\mathcal{N}} \sum_{k=1}^{\mathcal{N}} C_{n,k}^{(i,j),\text{inel}} f_k \Delta u_n \\
&= \sum_{i,j>i} \sum_{n=1}^{\mathcal{N}} \left[\sum_{k=m_n}^{m_n+1} C_{n,k}^{(i,j),\text{inel}} f_k \Delta u_n + \sum_{k=m_n}^{m_n+1} C_{k,k}^{(i,j),\text{inel}} f_k \Delta u_k \right] \\
&= \sum_{i,j>i} \sum_{n=1}^{\mathcal{N}} \left[\sum_{k=m_n}^{m_n+1} C_{n,k}^{(i,j),\text{inel}} f_k \Delta u_n - \sum_{k=m_n}^{m_n+1} C_{n,k}^{(i,j),\text{inel}} f_k \Delta u_n \right] \\
&= 0 \quad \text{with } u_{m_n}^{\text{node}} \leq u_n^{\text{node}} + V_{i,j} \quad ;
\end{aligned} \tag{64a}$$

$$\begin{aligned}
\int_0^\infty S_{\text{sup}}(u) du &= 0 \longrightarrow \sum_{n=1}^{\mathcal{N}} S_n^{\text{sup}} \Delta u_n = \sum_{n=1}^{\mathcal{N}} \sum_{i,j>i} S_{(i,j)_n}^{\text{sup}} \Delta u_n \\
&= \sum_{i,j>i} \sum_{n=1}^{\mathcal{N}} \sum_{k=1}^{\mathcal{N}} C_{n,k}^{(i,j),\text{sup}} f_k \Delta u_n \\
&= \sum_{i,j>i} \sum_{n=1}^{\mathcal{N}} \left[\sum_{k=m_n}^{m_n+\mathcal{M}+1} C_{n,k}^{(i,j),\text{sup}} f_k \Delta u_n + \sum_{k=m_n}^{m_n+\mathcal{M}+1} C_{k,k}^{(i,j),\text{sup}} f_k \Delta u_k \right] \\
&= \sum_{i,j>i} \sum_{n=1}^{\mathcal{N}} \left[\sum_{k=m_n}^{m_n+\mathcal{M}+1} C_{n,k}^{(i,j),\text{sup}} f_k \Delta u_n - \sum_{k=m_n}^{m_n+\mathcal{M}+1} C_{n,k}^{(i,j),\text{sup}} f_k \Delta u_n \right] \\
&= 0 \quad \text{with } u_{m_n}^{\text{node}} \leq u_n^{\text{node}} - V_{i,j} \quad .
\end{aligned} \tag{64b}$$

2.5 The power balance equation

The balance equation for the electron power-density, per electron at unit gas density, is obtained (depending on the particular PT/SST situation considered) by multiplying (2a), (3a) or (4a) by $u\sqrt{2e/m_e}$ and integrating it over all energies. The general result writes (in $\text{eV s}^{-1} \text{m}^3$)

$$\frac{\Theta(t)}{N} = \frac{\Theta_{\text{growth}}}{N} + \frac{\Theta_E}{N} + \frac{\Theta_{\text{coll}}}{N} , \quad (65a)$$

where the term in the left-hand side represents the intrinsic time-variation of the power density, and the different terms on the right-hand side represent, in order, the variation of the power density due to non-conservative mechanisms inducing a time/space net growth of the electron density, the power density gained from the applied electric field, and the power density gained/lost in collisional (“coll”) events, which can be separately written as (note that all “gain” terms are positively defined and all “loss” terms are negatively defined).

$$\frac{\Theta_{\text{coll}}}{N} \equiv \frac{\Theta_{\text{coll}}^{\text{gain}}}{N} + \frac{\Theta_{\text{coll}}^{\text{loss}}}{N} . \quad (65b)$$

The discretized form of (65a) can be obtained using the same finite-volume-like procedure as in section 2.2, applying Gauss’ theorem to integrals over one-dimensional energy volumes around each n -cell centered at energy u_n (see (22a)-(22b)).

The different terms in (65a) are written as follows.

- intrinsic time-variation term
The left-hand side of (65a) is

$$\frac{\Theta(t)}{N} = \begin{cases} 0 , & \text{for stationary simulations} \\ \frac{1}{N} \int_0^\infty u^{3/2} \frac{\partial f(u,t)}{\partial t} du = \frac{1}{N} \frac{d\varepsilon(t)}{dt} , & \text{for time-dependent simulations} \end{cases} . \quad (66)$$

The quantity $\Theta(t)/N$ is numerically calculated by summing all the power terms on the right-hand side of (65a).

- time/space net growth term

$$\frac{\Theta_{\text{growth}}}{N} = \begin{cases} -\frac{\langle \nu_{\text{eff}} \rangle}{N} \int_0^\infty \frac{1}{\sqrt{u}} u^2 f(u) du = -\frac{\langle \nu_{\text{eff}} \rangle}{N} \varepsilon \\ \rightarrow -\frac{\langle \nu_{\text{eff}} \rangle}{N} \sum_{n=1}^{\mathcal{N}} u_n^{3/2} f_n \Delta u_n , \\ \text{for a PT situation} \\ -\frac{\alpha_{\text{eff}}}{N} \frac{\gamma}{3} \int_0^\infty u^2 f^1(u) du = -\frac{\alpha_{\text{eff}}}{N} \frac{\gamma}{3} \int_0^\infty u^2 \left[-\frac{E/N}{\Omega_{\text{SST}}(u)} \frac{df(u)}{du} - \frac{\alpha_{\text{eff}}/N}{\Omega_{\text{SST}}(u)} f(u) \right] du \\ = -\frac{\alpha_{\text{eff}}}{N} (\mu_\varepsilon E - D_\varepsilon \alpha_{\text{eff}}) \\ \rightarrow -\frac{\alpha_{\text{eff}}}{N} \frac{\gamma}{3} \sum_{n=1}^{\mathcal{N}} \\ (u_n^{\text{node}})^2 \left\{ -\frac{E/N}{\Omega_{\text{SST}_n}^{\text{node}}} (f_n - f_{n-1}) - \frac{\alpha_{\text{eff}}/N}{\Omega_{\text{SST}_n}^{\text{node}}} [(1 - w_{n-1}) f_{n-1} + w_{n-1} f_n] \left(\frac{\Delta u_{n-1} + \Delta u_n}{2} \right) \right\} , \\ \text{for a SST situation (see equations (61)-(62))} \end{cases} \quad (67a)$$

Here, we have introduced the DC reduced transverse diffusion coefficient and the DC reduced mobility for energy transport (in $\text{eV m}^{-1} \text{s}^{-1}$ and $\text{m}^{-1} \text{s}^{-1}$, respectively)

$$D_\varepsilon N \equiv \frac{\gamma}{3} \int_0^\infty \frac{u^2}{\Omega_x(u)} f(u) du \quad (67b)$$

$$\mu_\varepsilon N \equiv -\frac{\gamma}{3} \int_0^\infty \frac{u^2}{\Omega_x(u)} \frac{df(u)}{du} du , \quad (67c)$$

with $x = c, \text{SST}$ for temporal or spatial growth of the electron density, respectively (see (5a)-(5b)).

- terms involving upflux functions $G_x(u)$

The $\Theta_{x=E,\text{coll}}/N$ terms involving upflux functions $G_x(u)$ are handled as follows

$$\begin{aligned}
\frac{\Theta_{x=E,\text{el,CAR}}}{N} &= -\frac{1}{N} \sum_n \int_{u_n^{\text{node}}}^{u_{n+1}^{\text{node}}} u \frac{dG_x(u)}{du} du = -\frac{1}{N} \sum_n u_n \left[G_{x_{n+1}}^{\text{node}} - G_{x_n}^{\text{node}} \right] \\
&= \gamma \left\{ \sum_{n=1}^{\mathcal{N}} u_n \left[g_{x_{n+1}}^{\text{node}} \left(d_x(1-w_n) - c_x \frac{2w_n}{\Delta u_n} \right) - g_{x_n}^{\text{node}} \left(d_x w_{n-1} + c_x \frac{2w_{n-1}}{\Delta u_{n-1}} \right) \right] f_n \right. \\
&\quad + \sum_{n=1}^{\mathcal{N}} u_n \left[g_{x_{n+1}}^{\text{node}} \left(d_x w_n + c_x \frac{2w_n}{\Delta u_n} \right) \right] f_{n+1} \\
&\quad \left. - \sum_{n=1}^{\mathcal{N}} u_n \left[g_{x_n}^{\text{node}} \left(d_x(1-w_{n-1}) - c_x \frac{2w_{n-1}}{\Delta u_{n-1}} \right) \right] f_{n-1} \right\} \\
&= \gamma \left\{ \sum_{n=1}^{\mathcal{N}} u_n \left[g_{x_{n+1}}^{\text{node}} \left(d_x(1-w_n) - c_x \frac{2w_n}{\Delta u_n} \right) - g_{x_n}^{\text{node}} \left(d_x w_{n-1} + c_x \frac{2w_{n-1}}{\Delta u_{n-1}} \right) \right] f_n \right. \\
&\quad + \sum_{n=2}^{\mathcal{N}+1} u_{n-1} \left[g_{x_n}^{\text{node}} \left(d_x w_{n-1} + c_x \frac{2w_{n-1}}{\Delta u_{n-1}} \right) \right] f_n \\
&\quad \left. - \sum_{n=0}^{\mathcal{N}-1} u_{n+1} \left[g_{x_{n+1}}^{\text{node}} \left(d_x(1-w_n) - c_x \frac{2w_n}{\Delta u_n} \right) \right] f_n \right\} \\
&= \gamma \left\{ \sum_{n=1}^{\mathcal{N}-1} (u_n - u_{n+1}) \left[g_{x_{n+1}}^{\text{node}} \left(d_x(1-w_n) - c_x \frac{2w_n}{\Delta u_n} \right) \right] f_n \right. \\
&\quad \left. - \sum_{n=2}^{\mathcal{N}} (u_n - u_{n-1}) \left[g_{x_n}^{\text{node}} \left(d_x w_{n-1} + c_x \frac{2w_{n-1}}{\Delta u_{n-1}} \right) \right] f_n \right\} \\
&= \gamma \sum_{n=1}^{\mathcal{N}} \left[g_{x_{n+1}}^{\text{node}} \left(c_x - d_x \frac{(1-w_n)\Delta u_n}{2w_n} \right) - g_{x_n}^{\text{node}} \left(c_x + d_x \frac{\Delta u_{n-1}}{2} \right) \right] f_n \quad , \quad (68a)
\end{aligned}$$

where we have used the boundary conditions $g_{x_1}^{\text{node}} = g_{x_{\mathcal{N}+1}}^{\text{node}} = 0$, and equation (19).

Therefore, the specific $\Theta_{x=E,\text{coll}}/N$ terms involving upflux functions $G_x(u)$ write

$$\begin{aligned}
\frac{\Theta_E}{N} &= -\frac{1}{N} \int_0^\infty u \frac{dG_E(u)}{du} du \\
&\rightarrow \gamma \sum_{n=1}^{\mathcal{N}} \left[g_{E_{n+1}}^{\text{node}} \left(c_E - d_E \frac{(1-w_n)\Delta u_n}{2w_n} \right) - g_{E_n}^{\text{node}} \left(c_E + d_E \frac{\Delta u_{n-1}}{2} \right) \right] f_n \quad (69a)
\end{aligned}$$

$$\begin{aligned}
\frac{\Theta_{x=\text{el,CAR}}}{N} &= -\frac{1}{N} \int_0^\infty u \frac{dG_x(u)}{du} du \\
&\rightarrow \gamma \sum_{n=1}^{\mathcal{N}} \left[g_{x_{n+1}}^{\text{node}} \left(c_x - d_x \frac{(1-w_n)\Delta u_n}{2w_n} \right) - g_{x_n}^{\text{node}} \left(c_x + d_x \frac{\Delta u_{n-1}}{2} \right) \right] f_n \quad (69b)
\end{aligned}$$

$$\begin{aligned}
\frac{\Theta_{x=\text{el,CAR}}^{\text{gain}}}{N} &= \gamma \int_0^\infty u \frac{d}{du} \left[g_x(u) c_x \frac{df(u)}{du} \right] du \\
&\rightarrow \gamma \sum_{n=1}^{\mathcal{N}} c_x \left[g_{x_{n+1}}^{\text{node}} - g_{x_n}^{\text{node}} \right] f_n \quad (69c)
\end{aligned}$$

$$\frac{\Theta_{x=\text{el,CAR}}^{\text{loss}}}{N} = \frac{\Theta_{x=\text{el,CAR}}}{N} - \frac{\Theta_{x=\text{el,CAR}}^{\text{gain}}}{N} \quad (69d)$$

$$\frac{\Theta_{ee}}{N} = -\frac{1}{N} \int_0^\infty u \frac{dG_{ee}(u)}{du} du$$

$$\begin{aligned}
& \rightarrow \gamma \sum_{n=1}^{\mathcal{N}} u_n [A_{n-1} f_{n-1} - (A_n + B_n) f_n + B_{n+1} f_{n+1}] \\
& = \gamma \left[\sum_{n=0}^{\mathcal{N}-1} u_{n+1} A_n f_n - \sum_{n=1}^{\mathcal{N}} u_n (A_n + B_n) f_n + \sum_{n=2}^{\mathcal{N}+1} u_{n-1} B_n f_n \right] \\
& = \gamma \left[\sum_{n=1}^{\mathcal{N}-1} (u_{n+1} - u_n) A_n f_n - \sum_{n=2}^{\mathcal{N}} (u_n - u_{n-1}) B_n f_n \right] \\
& = \gamma \sum_{n=1}^{\mathcal{N}} \left[\frac{\Delta u_n}{2w_n} A_n - \frac{\Delta u_{n-1}}{2w_{n-1}} B_n \right] f_n
\end{aligned} \tag{69e}$$

$$\begin{aligned}
\frac{\Theta_{ee}^{\text{gain}}}{N} & = \gamma \int_0^\infty u \frac{d}{du} \left[g_{ee}(u) J(u) \frac{df(u)}{du} \right] du \\
& \rightarrow \gamma \sum_{n=1}^{\mathcal{N}} \left[\frac{\Delta u_n}{2w_n} A_{J_n} - \frac{\Delta u_{n-1}}{2w_{n-1}} B_{J_n} \right] f_n
\end{aligned} \tag{69f}$$

$$\frac{\Theta_{ee}^{\text{loss}}}{N} = \frac{\Theta_{ee}}{N} - \frac{\Theta_{ee}^{\text{gain}}}{N} \tag{69g}$$

where we have used the boundary conditions $A_0 = A_{\mathcal{N}} = B_1 = B_{\mathcal{N}+1} = 0$, and equation (19).

Note that the power absorbed from the electric field can be written as (see (27a)-(27c))

- for a PT situation (see equations (27a)-(27c))

$$\begin{aligned}
\frac{\Theta_E}{N} & = \gamma \sum_{n=1}^{\mathcal{N}} \left[g_{E_{n+1}}^{\text{node}} - g_{E_n}^{\text{node}} \right] f_n \\
& = \gamma \left[\sum_{n=1}^{\mathcal{N}} g_{E_{n+1}}^{\text{node}} f_n - \sum_{n=1}^{\mathcal{N}} g_{E_n}^{\text{node}} f_n \right] \\
& = \gamma \left[\sum_{n=2}^{\mathcal{N}+1} g_{E_n}^{\text{node}} f_{n-1} - \sum_{n=1}^{\mathcal{N}} g_{E_n}^{\text{node}} f_n \right] \\
& = -\gamma \sum_{n=1}^{\mathcal{N}} g_{E_n}^{\text{node}} (f_n - f_{n-1}) \\
& = \left(\frac{E}{N} \right)^2 \left[-\frac{\gamma}{3} \sum_{n=1}^{\mathcal{N}} \frac{u_n^{\text{node}}}{\Omega_{\text{PT}_n}^{\text{node}}} (f_n - f_{n-1}) \right] \\
& = (\mu_e N) \left(\frac{E}{N} \right)^2,
\end{aligned} \tag{70a}$$

where we have used the boundary conditions $g_{E_1}^{\text{node}} = g_{E_{\mathcal{N}+1}}^{\text{node}} = 0$, and equation (62).

- for a SST situation (see equations (27a)-(27c))

$$\begin{aligned}
\frac{\Theta_E}{N} & = \gamma \sum_{n=1}^{\mathcal{N}} \frac{E/N}{\alpha_{\text{eff}}/N} \left[g_{E_{n+1}}^{\text{node}} - g_{E_n}^{\text{node}} \right] f_n - \gamma \sum_{n=1}^{\mathcal{N}} \left[g_{E_{n+1}}^{\text{node}} \frac{(1-w_n)\Delta u_n}{2w_n} + g_{E_n}^{\text{node}} \frac{\Delta u_{n-1}}{2} \right] f_n \\
& = \gamma \left[\sum_{n=1}^{\mathcal{N}} \frac{E/N}{\alpha_{\text{eff}}/N} g_{E_{n+1}}^{\text{node}} f_n - \sum_{n=1}^{\mathcal{N}} \frac{E/N}{\alpha_{\text{eff}}/N} g_{E_n}^{\text{node}} f_n - \sum_{n=1}^{\mathcal{N}} g_{E_{n+1}}^{\text{node}} \frac{(1-w_n)\Delta u_n}{2w_n} f_n - \sum_{n=1}^{\mathcal{N}} g_{E_n}^{\text{node}} \frac{\Delta u_{n-1}}{2} f_n \right] \\
& = \gamma \left[\sum_{n=2}^{\mathcal{N}+1} \frac{E/N}{\alpha_{\text{eff}}/N} g_{E_n}^{\text{node}} f_{n-1} - \sum_{n=1}^{\mathcal{N}} \frac{E/N}{\alpha_{\text{eff}}/N} g_{E_n}^{\text{node}} f_n \right. \\
& \quad \left. - \sum_{n=2}^{\mathcal{N}+1} g_{E_n}^{\text{node}} \frac{(1-w_{n-1})\Delta u_{n-1}}{2w_{n-1}} f_{n-1} - \sum_{n=1}^{\mathcal{N}} g_{E_n}^{\text{node}} \frac{\Delta u_{n-1}}{2} f_n \right]
\end{aligned}$$

$$\begin{aligned}
&= -\gamma \frac{E/N}{\alpha_{\text{eff}}/N} \sum_{n=1}^{\mathcal{N}} g_{E_n}^{\text{node}} (f_n - f_{n-1}) - \gamma \sum_{n=1}^{\mathcal{N}} g_{E_n}^{\text{node}} [(1 - w_{n-1})f_{n-1} + w_{n-1}f_n] \frac{\Delta u_{n-1}}{2w_{n-1}} \\
&= -\gamma \frac{E/N}{\alpha_{\text{eff}}/N} \sum_{n=1}^{\mathcal{N}} g_{E_n}^{\text{node}} (f_n - f_{n-1}) \\
&\quad - \gamma \sum_{n=1}^{\mathcal{N}} g_{E_n}^{\text{node}} [(1 - w_{n-1})f_{n-1} + w_{n-1}f_n] \left(\frac{\Delta u_{n-1} + \Delta u_n}{2} \right) \\
&= \left(\frac{E}{N} \right)^2 \left[-\frac{\gamma}{3} \sum_{n=1}^{\mathcal{N}} \frac{u_n^{\text{node}}}{\Omega_{\text{SST}_n}^{\text{node}}} (f_n - f_{n-1}) \right] \\
&\quad - \frac{E}{N} \frac{\alpha_{\text{eff}}}{N} \left[\frac{\gamma}{3} \sum_{n=1}^{\mathcal{N}} \frac{u_n^{\text{node}}}{\Omega_{\text{SST}_n}^{\text{node}}} [(1 - w_{n-1})f_{n-1} + w_{n-1}f_n] \left(\frac{\Delta u_{n-1} + \Delta u_n}{2} \right) \right] \\
&= (\mu_e N) \left(\frac{E}{N} \right)^2 - (D_e N) \frac{E}{N} \frac{\alpha_{\text{eff}}}{N} , \tag{70b}
\end{aligned}$$

where we have used the boundary conditions $g_{E_1}^{\text{node}} = g_{E_{\mathcal{N}+1}}^{\text{node}} = 0$, and equations (61), (62) and (17c).

- inelastic (loss) terms (see equations (39a)-(39c))

$$\begin{aligned}
\frac{\Theta_{\text{inel}}^{\text{loss}}}{N} &= \sum_{i,j>i} \frac{\Theta_{(i,j),\text{inel}}^{\text{loss}}}{N} \\
&= \sum_{i,j>i} \gamma \int_0^\infty S_{(i,j),\text{inel}}(u) u \, du \\
&= \sum_{i,j>i} \gamma \int_0^\infty \left[S_{(i,j),\text{inel}}^{(+)}(u) + S_{(i,j),\text{inel}}^{(-)}(u) \right] u \, du \tag{71a}
\end{aligned}$$

and

$$\begin{aligned}
\frac{\Theta_{(i,j),\text{inel}}^{\text{loss}}}{N} &= \gamma \sum_{n=1}^{\mathcal{N}} \int_{u_n^{\text{node}}}^{u_{n+1}^{\text{node}}} \left[S_{(i,j),\text{inel}}^{(+)}(u) + S_{(i,j),\text{inel}}^{(-)}(u) \right] u \, du \\
&= \gamma \sum_{n=1}^{\mathcal{N}} \sum_{k=1}^{\mathcal{N}} C_{n,k}^{(i,j),\text{inel}} f_k u_n \Delta u_n \\
&= \gamma \sum_{n=1}^{\mathcal{N}} \left[\sum_{k=m_n}^{m_n+1} C_{n,k}^{(i,j),\text{inel}} f_k u_n \Delta u_n + \sum_{k=m_n}^{m_n+1} C_{k,k}^{(i,j),\text{inel}} f_k u_k \Delta u_k \right] \\
&= \gamma \sum_{n=1}^{\mathcal{N}} \left[\sum_{k=m_n}^{m_n+1} C_{n,k}^{(i,j),\text{inel}} f_k u_n \Delta u_n - \sum_{k=m_n}^{m_n+1} C_{n,k}^{(i,j),\text{inel}} f_k u_k \Delta u_n \right] \\
&= -\gamma \sum_{n=1}^{\mathcal{N}} \sum_{k=m_n}^{m_n+1} C_{n,k}^{(i,j),\text{inel}} f_k (u_k - u_n) \Delta u_n
\end{aligned}$$

$$= \begin{cases} -\gamma\delta_i \sum_{n=1}^{\mathcal{N}} \sigma_{(i,j)_{m_n}} f_{m_n} \left[\frac{1}{2}(u_n^{\text{node}} + u_{n+1}^{\text{node}}) + V_{i,j} \right] (u_{n+1}^{\text{node}} - u_n^{\text{node}})(u_{m_n} - u_n) \\ \quad \text{if } u_{m_n}^{\text{node}} \leq u_n^{\text{node}} + V_{i,j}, u_{n+1}^{\text{node}} + V_{i,j} \leq u_{m_n+1}^{\text{node}} \\ \\ -\gamma\delta_i \sum_{n=1}^{\mathcal{N}} \left\{ \sigma_{(i,j)_{m_n}} f_{m_n} \frac{1}{2}(u_n^{\text{node}} + V_{i,j} + u_{m_n+1}^{\text{node}}) [(u_{m_n+1}^{\text{node}} - V_{i,j}) - u_n^{\text{node}}] (u_{m_n} - u_n) + \right. \\ \quad \left. \sigma_{(i,j)_{m_n+1}} f_{m_n+1} \frac{1}{2}(u_{n+1}^{\text{node}} + V_{i,j} + u_{m_n+1}^{\text{node}}) [u_{n+1}^{\text{node}} - (u_{m_n+1}^{\text{node}} - V_{i,j})] (u_{m_n+1} - u_n) \right\} \\ \quad \text{if } u_{m_n}^{\text{node}} < u_n^{\text{node}} + V_{i,j} < u_{m_n+1}^{\text{node}} < u_{n+1}^{\text{node}} + V_{i,j} < u_{m_n+2}^{\text{node}} \end{cases} \quad (71b)$$

Note that the power dissipated in the inelastic excitation $i \rightarrow j$ is analytically evaluated as (see (37))

$$\begin{aligned} \frac{\Theta_{(i,j),\text{inel}}^{\text{loss}}}{N} &= \gamma \int_0^\infty \left[S_{(i,j),\text{inel}}^{(+)}(u) + S_{(i,j),\text{inel}}^{(-)}(u) \right] u du \\ &= \gamma \int_0^\infty [\delta_i(u + V_{i,j})\sigma_{i,j}(u + V_{i,j})f(u + V_{i,j}) - \delta_i u \sigma_{i,j}(u)f(u)] u du \\ &= \gamma\delta_i \int_{V_{i,j}}^\infty (u' - V_{i,j})u'\sigma_{i,j}(u')f(u')du' - \gamma\delta_i \int_0^\infty u^2\sigma_{i,j}(u)f(u)du \\ &= \gamma\delta_i \int_0^\infty u'^2\sigma_{i,j}(u')f(u')du' - \gamma\delta_i \int_0^\infty V_{i,j}u'\sigma_{i,j}(u')f(u')du' - \gamma\delta_i \int_0^\infty u^2\sigma_{i,j}(u)f(u)du \\ &= -\delta_i V_{i,j} \gamma \int_0^\infty u\sigma_{i,j}(u)f(u)du \\ &= -\delta_i C_{i,j} V_{i,j} \quad , \end{aligned} \quad (72a)$$

where again we have considered that $\sigma_{i,j}(u) = 0$ for $u \leq V_{i,j}$, and where we have introduced the rate coefficient for the inelastic excitation $i \rightarrow j$

$$C_{i,j} \equiv \gamma \int_0^\infty u\sigma_{i,j}(u)f(u)du \quad . \quad (72b)$$

In general, the discretized form (71b) of the power dissipated in inelastic collisions doesn't allow retrieving the corresponding continuous form (72a). The exception is the case of an uniform energy-grid, for which we have

$$\begin{aligned} \frac{\Theta_{(i,j),\text{inel}}^{\text{loss}}}{N} &= -\gamma\delta_i \sum_{n=1}^{\mathcal{N}} \sigma_{(i,j)_{m_n}} f_{m_n} \left[\frac{1}{2}(u_n^{\text{node}} + u_{n+1}^{\text{node}}) + V_{i,j} \right] (u_{n+1}^{\text{node}} - u_n^{\text{node}})(u_{m_n} - u_n) \\ &= -\gamma\delta_i \sum_{n=1}^{\mathcal{N}} \sigma_{(i,j)_{m_n}} f_{m_n} (u_n + V_{i,j})(u_{m_n} - u_n) \Delta u_n \\ &= -\gamma\delta_i \sum_{n=1}^{\mathcal{N}} \sigma_{(i,j)_{n+l_{i,j}}} f_{n+l_{i,j}} u_{n+l_{i,j}} V_{i,j} \Delta u_n \\ &= -V_{i,j} \delta_i \gamma \sum_{n=1+l_{i,j}}^{\mathcal{N}+l_{i,j}} \sigma_{(i,j)_n} f_n u_n \Delta u_n \\ &= -V_{i,j} \delta_i \gamma \sum_{n=1}^{\mathcal{N}} \sigma_{(i,j)_n} f_n u_n \Delta u_n \\ &= -\delta_i C_{i,j} V_{i,j} \quad , \end{aligned} \quad (73a)$$

with $\sigma_{(i,j)_n} = 0$ for $n = 1, \dots, l_{i,j}$ and $n > \mathcal{N}$, and

$$C_{i,j} = \gamma \sum_{n=1}^{\mathcal{N}} \sigma_{(i,j)_n} f_n u_n \Delta u_n \quad (73b)$$

corresponding to the discretized form of (72b).

The numerical difference between the results of (71b) and (72a), using the discretized rate coefficient (73b), will provide information on the quality of the discretization and should take part of the criteria to define / adjust the non-uniform energy grid.

- superelastic (gain) terms (see equations (42a)-(42c))

$$\begin{aligned}
\frac{\Theta_{\text{sup}}^{\text{gain}}}{N} &= \sum_{i,j>i} \frac{\Theta_{(i,j),\text{sup}}^{\text{gain}}}{N} \\
&= \sum_{i,j>i} \gamma \int_0^\infty S_{(i,j),\text{sup}}(u) u \, du \\
&= \sum_{i,j>i} \gamma \int_0^\infty \left[S_{(i,j),\text{sup}}^{(+)}(u) + S_{(i,j),\text{sup}}^{(-)}(u) \right] u \, du
\end{aligned} \tag{74a}$$

and

$$\begin{aligned}
\frac{\Theta_{(i,j),\text{sup}}^{\text{gain}}}{N} &= \gamma \sum_{n=1}^{\mathcal{N}} \int_{u_n^{\text{node}}}^{u_{n+1}^{\text{node}}} \left[S_{(i,j),\text{sup}}^{(+)}(u) + S_{(i,j),\text{sup}}^{(-)}(u) \right] u \, du \\
&= \gamma \sum_{n=1}^{\mathcal{N}} \sum_{k=1}^{\mathcal{N}} C_{n,k}^{(i,j),\text{sup}} f_k u_n \Delta u_n \\
&= \gamma \sum_{n=1}^{\mathcal{N}} \left[\sum_{k=m_n}^{m_n+\mathcal{M}+1} C_{n,k}^{(i,j),\text{sup}} f_k u_n \Delta u_n + \sum_{k=m_n}^{m_n+\mathcal{M}+1} C_{k,k}^{(i,j),\text{sup}} f_k u_k \Delta u_k \right] \\
&= \gamma \sum_{n=1}^{\mathcal{N}} \left[\sum_{k=m_n}^{m_n+\mathcal{M}+1} C_{n,k}^{(i,j),\text{sup}} f_k u_n \Delta u_n - \sum_{k=m_n}^{m_n+\mathcal{M}+1} C_{n,k}^{(i,j),\text{sup}} f_k u_k \Delta u_n \right] \\
&= \gamma \sum_{n=1}^{\mathcal{N}} \sum_{k=m_n}^{m_n+\mathcal{M}+1} C_{n,k}^{(i,j),\text{sup}} f_k (u_n - u_k) \Delta u_n \\
&= \gamma \delta_j \frac{g_i}{g_j} \sum_{n=1}^{\mathcal{N}} \left\{ \sigma_{(i,j)_n} f_{m_n} \left[\frac{1}{2} [(u_n^{\text{node}} - V_{i,j}) + u_{m_n+1}^{\text{node}}] + V_{i,j} \right] [u_{m_n+1}^{\text{node}} - (u_n^{\text{node}} - V_{i,j})] (u_n - u_{m_n}) + \right. \\
&\quad (1 - \delta_{\mathcal{M},0}) \sum_{p=1}^{\mathcal{M}} \sigma_{(i,j)_n} f_{m_n+p} \left[\frac{1}{2} (u_{m_n+p}^{\text{node}} + u_{m_n+p+1}^{\text{node}}) + V_{i,j} \right] (u_{m_n+p+1}^{\text{node}} - u_{m_n+p}^{\text{node}}) (u_n - u_{m_n+p}) + \\
&\quad \left. \sigma_{(i,j)_n} f_{m_n+\mathcal{M}+1} \left[\frac{1}{2} [u_{m_n+\mathcal{M}+1} + (u_{n+1}^{\text{node}} - V_{i,j})] + V_{i,j} \right] [(u_{n+1}^{\text{node}} - V_{i,j}) - u_{m_n+\mathcal{M}+1}^{\text{node}}] \times \right. \\
&\quad \left. (u_n - u_{m_n+\mathcal{M}+1}) \right\} ,
\end{aligned} \tag{74b}$$

for $u_{m_n}^{\text{node}} \leq u_n^{\text{node}} - V_{i,j}$.

Note that the power dissipated in the superelastic deexcitation $j \rightarrow i$ is analytically evaluated as (see (40))

$$\begin{aligned}
\frac{\Theta_{(i,j),\text{sup}}^{\text{gain}}}{N} &= \gamma \int_0^\infty \left[S_{(i,j),\text{sup}}^{(+)}(u) + S_{(i,j),\text{sup}}^{(-)}(u) \right] u \, du \\
&= \gamma \int_0^\infty \left[\delta_j \frac{g_i}{g_j} u \sigma_{i,j}(u) f(u - V_{i,j}) - \delta_j \frac{g_i}{g_j} (u + V_{i,j}) \sigma_{i,j}(u + V_{i,j}) f(u) \right] u \, du \\
&= \gamma \delta_j \frac{g_i}{g_j} \left[\int_{V_{i,j}}^\infty u^2 \sigma_{i,j}(u) f(u - V_{i,j}) du - \int_{V_{i,j}}^\infty u' (u' - V_{i,j}) \sigma_{i,j}(u') f(u' - V_{i,j}) du' \right]
\end{aligned}$$

$$\begin{aligned}
&= \gamma \delta_j \frac{g_i}{g_j} \left[\int_{V_{i,j}}^{\infty} u^2 \sigma_{i,j}(u) f(u - V_{i,j}) du - \int_{V_{i,j}}^{\infty} u'^2 \sigma_{i,j}(u') f(u' - V_{i,j}) du' + \right. \\
&\quad \left. \int_{V_{i,j}}^{\infty} u' V_{i,j} \sigma_{i,j}(u') f(u' - V_{i,j}) du' \right] \\
&= \gamma \delta_j \frac{g_i}{g_j} V_{i,j} \int_{V_{i,j}}^{\infty} u \sigma_{i,j}(u) f(u - V_{i,j}) du \\
&= \gamma \delta_j \frac{g_i}{g_j} V_{i,j} \int_0^{\infty} (u + V_{i,j}) \sigma_{i,j}(u + V_{i,j}) f(u) du \\
&= \delta_j V_{i,j} \gamma \int_0^{\infty} u \sigma_{j,i}(u) f(u) du \\
&= \delta_j C_{j,i} V_{i,j} \quad , \tag{75a}
\end{aligned}$$

where again we have considered that $\sigma_{i,j}(u) = 0$ for $u \leq V_{i,j}$, and where we have used the Klein-Rosseland relation and introduced the rate coefficient for the superelastic deexcitation $j \rightarrow i$

$$C_{j,i} \equiv \gamma \int_0^{\infty} u \sigma_{j,i}(u) f(u) du = \frac{g_i}{g_j} \gamma \int_0^{\infty} (u + V_{i,j}) \sigma_{i,j}(u + V_{i,j}) f(u) du \quad . \tag{75b}$$

Similarly to the inelastic term, the discretized form (74b) of the power gained in superelastic collisions doesn't allow retrieving the corresponding continuous form (75a). The exception is the case of an uniform energy-grid, for which we have

$$\begin{aligned}
\frac{\Theta_{(i,j),\text{sup}}^{\text{gain}}}{N} &= \gamma \delta_j \frac{g_i}{g_j} \sum_{n=1}^{\mathcal{N}} \sigma_{(i,j)_n} f_{m_n} \left[\frac{1}{2} [(u_n^{\text{node}} - V_{i,j}) + u_{m_n+1}^{\text{node}}] + V_{i,j} \right] \times \\
&\quad [u_{m_n+1}^{\text{node}} - (u_n^{\text{node}} - V_{i,j})] (u_n - u_{m_n}) \\
&= \gamma \delta_j \frac{g_i}{g_j} \sum_{n=1}^{\mathcal{N}} \sigma_{(i,j)_n} f_{m_n} \left[\frac{1}{2} [(u_n^{\text{node}} - V_{i,j}) + (u_{n+1}^{\text{node}} - V_{i,j})] + V_{i,j} \right] \times \\
&\quad [(u_{n+1}^{\text{node}} - V_{i,j}) - (u_n^{\text{node}} - V_{i,j})] (u_n - u_{m_n}) \\
&= \gamma \delta_j \frac{g_i}{g_j} \sum_{n=1}^{\mathcal{N}} \sigma_{(i,j)_n} f_{m_n} u_n (u_n - u_{m_n}) \Delta u_n \\
&= \gamma \delta_j \frac{g_i}{g_j} \sum_{n=1}^{\mathcal{N}} \sigma_{(i,j)_n} f_{n-l_{i,j}} u_n V_{i,j} \Delta u_n \\
&= V_{i,j} \delta_j \frac{g_i}{g_j} \gamma \sum_{n=1-l_{i,j}}^{\mathcal{N}-l_{i,j}} \sigma_{(i,j)_{n+l_{i,j}}} f_n u_{n+l_{i,j}} \Delta u_n \\
&= V_{i,j} \delta_j \frac{g_i}{g_j} \gamma \sum_{n=1}^{\mathcal{N}} \sigma_{(i,j)_{n+l_{i,j}}} f_n u_{n+l_{i,j}} \Delta u_n \\
&= \delta_j C_{j,i} V_{i,j} \quad , \tag{76a}
\end{aligned}$$

with $\sigma_{(i,j)_n} = 0$ for $n = 1, \dots, l_{i,j}$ and $n > \mathcal{N}$, and

$$C_{j,i} = \gamma \frac{g_i}{g_j} \sum_{n=1}^{\mathcal{N}} \sigma_{(i,j)_{n+l_{i,j}}} f_n u_{n+l_{i,j}} \Delta u_n \tag{76b}$$

corresponding to the discretized form of (75b).

Similarly to the inelastic operator, the numerical difference between the results of (74b) and (75a), using the discretized rate coefficient (76b), will provide information on the quality of the discretization and should take part of the criteria to define / adjust the non-uniform energy grid.

- TBC

$$\begin{aligned}
\frac{\Theta_{\text{ion}}}{N} &= \gamma \sum_i \int_0^\infty S_{i,\text{ion}}(u) u du = - \sum_i \delta_i C_{i,\text{ion}} V_{i,\text{ion}} \\
&\rightarrow \gamma \sum_i \sum_{n=1}^{\mathcal{N}} S_{(i,\text{ion})_n} u_n \Delta u
\end{aligned} \tag{77a}$$

$$\begin{aligned}
\frac{\Theta_{\text{att}}}{N} &= \gamma \sum_i \int_0^\infty S_{i,\text{att}}(u) u du \\
&\rightarrow \gamma \sum_i \sum_{n=1}^{\mathcal{N}} S_{(i,\text{att})_n} u_n \Delta u
\end{aligned} \tag{77b}$$